Policing, Protection, and Participation: The Effects of a Randomized 'Hotspots' Experiment on Voter Turnout

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Abstract

Existing scholarship links involuntary police contact to political withdrawal, but many high-crime, disadvantaged communities suffer from a shortage of police protection in addition to its excesses. Whether policing that responds to this insecurity mobilizes or demobilizes these communities, and whether these responses depend on how officers police, is not clear. I examine these questions by linking individual voter records to the Jacksonville Hotspots Experiment, which randomized police deployments and tactics across high-crime neighborhoods. Some neighborhoods received high-intensity patrols that generated many stops and arrests; others received patrols emphasizing non-enforcement, community-oriented tactics; while a control group experienced no changes. Both the high- and low-enforcement interventions raised turnout in subsequent elections by 2 to 4 percentage points, concentrated in the highest-crime areas. The results suggest that visible police responses to community violence can mobilize political engagement, complicating accounts that center punitive contact as the primary way policing shapes political life.

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Introduction

Scholars have long observed that marginalized communities in the United States experience both punitive over-policing and inadequate protection from crime and violence. Low rates of political participation in poor, racially segregated neighborhoods with high crime rates are often attributed to both of these failures. Despite the prominence of this account, there is surprisingly little empirical evidence on the political consequences of policies meant to remedy under-policing in poor communities. Can large infusions of police patrols increase turnout in communities with high crime rates? And importantly, does this political response depend on *how* officers police these places?

Whether such patrols might mobilize or demobilize members of these communities is not obvious. The presence of the police may raise turnout by reducing crime, reassuring the public, and signaling government responsiveness. The same deployments, however, expose the public to more stops, citations, and arrests—the kinds of involuntary encounters that a large body of work links to political withdrawal (Lerman and Weaver 2020; Ben-Menachem and Morris 2023). And in neighborhoods with long histories of adversarial policing, sudden surges in patrols, however well-intentioned, may be viewed as a form of surveillance rather than protection.

Existing evidence offers limited purchase on these questions. Most work on the political effects of policing in the United States examines forms of police coercion and violence that are rare, extreme, or unambiguously harmful, such as being involuntarily stopped, arrested, and incarcerated, or spatial proximity to fatal shootings and egregious cases of racial profiling (e.g. Ang and Tebes 2024; White 2019; Naftel, *Forthcoming*). Officers spend the vast majority of their time on patrol and responding to calls for service, and we know far less about the political consequences of that routine visibility, including whether these patrols might encourage civic and political engagement by providing security. The work that does take up this question relies on rough proxies such as quasi-random variation in municipal police hiring (Romero 2025), which cannot disentangle the political effects of where police are deployed from *how* they police, or speak to policing's effects in high-crime, disadvantaged communities specifically. Other work focuses on counter-insurgency and security provision in post-conflict societies (e.g. Blair, Karim, and Morse 2019; Condra et al. 2019) — settings far removed from routine police work.

I study these questions through a secondary analysis of a large-scale, randomized police experiment conducted in Jacksonville, Florida, by the Police Executive Research Forum (Taylor, Koper, and Woods 2011). Between January and April 2009, the Jacksonville Sheriff's Office and PERF researchers identified 83 hotspots of violent crime — overwhelmingly located in poor, predominantly Black neighborhoods — and randomly

assigned them to one of two treatments or a control group. Half of the treated hotspots were saturated with police patrols for an average of 53 hours per week, which resulted in an 85% increase in investigatory stops and arrests during the study period (Taylor, Koper, and Woods 2011). The other half also received large increases in police patrols; however, officers in this treatment arm were trained to use community-oriented problem solving approaches to reduce violence, and encouraged to avoid enforcement whenever possible. Arrest data and activity logs confirm that these officers were far less likely to rely on coercion. Instead, officers conducted surveys of residents and coordinated with property owners and other city agencies to improve infrastructure and access to social services (Taylor, Koper, and Woods 2011). These randomized changes in police presence and tactics present a unique opportunity to study the effects of police surges in communities with high crime rates.

Linking individual voter files and local election returns to the experimental hotspots, I find that both interventions raised voter turnout in subsequent elections by 2 to 4 percentage points, with the clearest effects among Black voters. Strikingly, this mobilizing effect is statistically indistinguishable across the two treatment arms, despite large differences in the propensity of officers to engage in stops and arrests. And while these gains are concentrated in the hotspots with the highest pre-treatment rates of violence, neither intervention produced durable reductions in crime, making it difficult to simply attribute the turnout gains to lower levels of victimization. These turnout increases also did not translate into electoral rewards for incumbents, including the incumbent sheriff who oversaw the program. Drawing on local news coverage and police computer-aided dispatch data, I find no evidence of backlash, such as protests or spikes in fear-driven calls. Instead, I argue that the evidence is most consistent with an interpretation in which a visible police response to neighborhood violence reassured residents concerned with crime, in line with prior findings that visible patrols can reduce fear and improve evaluations of the police (Innes 2004).

These findings provide multiple contributions to the literature on policing and political behavior. First, this setting allows me to study forms of policing that are typically unobserved. Most police administrative data record coercive events (stops, citations, arrests) but provide little systematic measurement of community-oriented police activities of the type used in Jacksonville, making their political consequences difficult to estimate. Second, the randomized, place-based design overcomes common challenges in the study of the community-level effects of policing by defining a clear treated population and yielding causal estimates that include members of the public who were never directly contacted by the police. Third, by directly comparing two types of interventions that differed sharply in their approaches to crime control, the findings speak to which dimensions of policing the public responds to. The results suggest that the aggregate participatory effects of policing are not well captured by counts of coercive encounters alone, and that the service functions

of law enforcement may be at least as important as punitive contact in shaping electoral behavior.

The findings also have implications for theories of policy feedback and street-level bureaucracy. A central claim in this tradition is that citizens draw political lessons from the state through the quality and character of their interactions with front-line agents, whose behavior sends messages about standing and government responsiveness (Soss 1999; Lipsky 2010). This experiment would seem an especially likely setting for those differences to matter, given the substantial differences in officer selection, training, and behavior it generated between the two interventions. One interpretation of the similar turnout effects across arms is that, for the many individuals who never experienced direct contact with officers, the most salient information produced by these interventions was not how officers interacted with the public but the visible signs that the police were investing resources in their neighborhood. More broadly, the results highlight how the features of policies that drive feedback effects among the "policy adjacent" (Hankinson, Magazinnik, and Sands 2026) can be quite different from the features that matter to the direct targets of policies.

Finally, the results contribute to the broader study of policing's social consequences. A large experimental literature in criminology has assessed the impacts of proactive, place-based police strategies and generally concludes that these interventions yield modest but positive effects in high-crime communities (e.g. National Academies 2018). But this work is often narrowly focused on outcomes such as views of the police and crime, and relies heavily on resident surveys with low response rates and substantial attrition. By leveraging administrative voter files, I show that proactive policing strategies also affect *political* behavior, illustrating how increasingly granular administrative records can be used to revisit these field experiments and extend them beyond the outcomes they were designed to study.

Theoretical Background

Policing As a Source of Negative Political Learning

When police departments deploy more personnel to a neighborhood, they often increase the probability that residents of those places will be stopped, ticketed, searched, or arrested. A large body of work finds that these involuntary encounters can demobilize voters (Ben-Menachem and Morris 2023; White 2019). This is thought to be due to the material costs of criminal justice involvement — which can range from license suspensions to loss of employment and heavy fines — as well as the civic lessons these encounters transmit. These experiences may signal that at best, the government is cruel and unresponsive, and at worst, that the government is something to be actively avoided (Lerman and Weaver 2020).

Importantly, the political consequences of policing are not confined to those who experience direct contact; they spill outward through neighborhood networks to shape the attitudes and behavior of the broader community (Burch 2013). Members of the public often learn about stops, searches, and arrests vicariously from relatives, friends, and media, and these second-hand accounts color expectations about their own prospects of being targeted, the fairness of the legal process, and the state's willingness to protect them (Anoll, Epp, and Israel-Trummel 2022). Because involuntary encounters are often densely concentrated in poor, predominantly Black neighborhoods, these vicarious experiences can contribute to a shared understanding of the police as predatory and ineffective, and the law as selectively enforced (Bell 2017; Kirk and Papachristos 2011). As with direct contact, this "legal cynicism" is often thought to erode civic engagement by encouraging the avoidance of institutions and lowering the perceived returns of electoral participation (Laniyonu 2019; Lerman and Weaver 2020). But in rare cases — typically after highly publicized violence or abuse — the same shared understandings can catalyze collective action, prompting protests and electoral mobilization among those who feel targeted on the basis of race or class (Ang and Tebes 2024; Naftel, *Forthcoming*; Garcia-Rios et al. 2023). Together, this suggests that policing can mobilize or demobilize voters via a variety of harm-centered mechanisms, including material burdens, fear, anger, or a sense of injustice. However, this literature's theoretical and empirical focus on the electoral effects of policing's harms (see Appendix Table B.1 for a review) may understate the feedback effects of police service provision, particularly in communities with high crime rates.

The Consequences of Under-Policing

Many of the same neighborhoods that experience frequent police contact also face a deficit of basic protection from violence. This is particularly true in poor, predominantly Black communities, where many express a "dual frustration" with law enforcement, condemning police harassment and the uneven application of the law while also faulting officers for failing to keep them safe (Carr, Napolitano, and Keating 2007; Meares 1997).

This deficit of basic security carries political consequences of its own. While isolated instances of criminal victimization can mobilize voters (Bateson 2012), a growing body of work suggests that chronic exposure to violence and other forms of criminal activity imposes substantial social, economic, and psychological harms that can discourage political engagement, particularly among Black Americans (Moffett-Bateau 2023; Jones, *Forthcoming*). Exposure to crime, as well as direct victimization, has been linked to chronic stress and social isolation (Cornaglia, Feldman, and Leigh 2014; Dustmann and Fasani 2016; Ross, Mirowsky, and Pribesh 2001; Moffett-Bateau 2023), reducing the resources and opportunities individuals have at their disposal to

become involved with politics. Beyond these material harms, the failure of the police to provide security may lead individuals to understand law enforcement as indifferent, a judgment thought to generalize to other government institutions and discourage political engagement (Jones, *Forthcoming*).

Given this, we may expect policing interventions that alleviate these burdens to raise turnout. A substantial body of evidence finds that visible patrols deter crime and reduce visible signs of disorder (Braga et al. 2019), and may also signal to the public that the state is committed to addressing neighborhood problems (Innes 2004). Policing interventions in high-crime communities may also reshape neighborhood social dynamics in ways that encourage participation. Studying a randomized hotspots experiment in St. Louis County, Kochel and Weisburd (2019) find that an increase in police patrols raised individuals' sense of closeness with their neighbors and willingness to intervene against social disorder, suggesting that greater perceived police reliability can embolden residents of high-crime neighborhoods to solve collective problems and enforce social norms. Improved trust and a stronger sense of collective efficacy could in turn lead to greater civic engagement. Ties with neighbors can induce social pressure to participate in politics, facilitate the diffusion of electoral information, and provide opportunities to become involved with campaigns or community projects (McClurg 2003; Sinclair 2012; Verba, Schlozman, and Brady 1995). A climate of informal social control also reduces the personal risks (real or perceived) of public activity in high-crime areas, making it easier to attend meetings, host registration drives, or simply walk to the polls after dark.

Yet there are also reasons to expect that such interventions may fail to translate into turnout or even backfire. Any participatory gains from improved safety may be offset if intensified patrols simultaneously increase stops and arrests, given the widespread assumption that these coercive encounters discourage political participation. And while criminologists have found that some forms of policing interventions can improve collective efficacy and views of the police, it is not obvious that these outcomes should translate directly into political participation. Measures of collective efficacy used by criminologists and sociologists are reliable predictors of neighborhood crime rates (Sampson, Raudenbush, and Earls 1997), but their relationship to political behavior has received scant empirical attention (Levine et al. 2018). And while recent experimental work finds that positive, non-enforcement encounters with law enforcement can improve views of the police (Peyton, Sierra-Arévalo, and Rand 2019; Karim 2020), these encounters had no detectable effect on views of government more broadly. This mirrors other recent work finding that direct and visible government benefits tend to raise support for those specific programs without shifting citizens' broader orientations toward the state (Anzia, Jares, and Malhotra 2022).

Whether interventions meant to address issues of under-policing might increase participation in poor neighborhoods with high crime rates remains an open question. Most evidence on the political feedback effects

of the criminal justice system comes from direct, involuntary encounters with the police and courts, rather than the net effect of policing on communities as a whole. Evidence for the community-level effects of security provision comes largely from fragile or post-conflict states, where interventions have produced null or negative effects on participation and trust (Condra et al. 2019; Blair, Karim, and Morse 2019). Within the United States, neighborhood-level work linking police activity to turnout rests on long-term, cross-sectional differences (e.g. Laniyonu 2019) that are confounded by the endogeneity of policing, poverty, and crime. And while Romero (2025) finds that plausibly exogenous expansions of local police forces increased voting among Black men, this hiring-based variation tells us little about exposure to policing activity in the communities at the center of under-policing accounts. It also remains unclear whether these political responses depend on how officers go about reducing crime, a question I turn to next.

How Much do Police Tactics Matter?

Influential traditions in the literatures on the carceral state (Anoll, Epp, and Israel-Trummel 2022; Lerman and Weaver 2014) and procedural justice (National Academies 2018) assume that the public's reactions to the police depend on the character of officers' conduct. In particular, scholars have focused on the political impacts of officers' propensity to make stops and arrests (e.g. Laniyonu 2019; Palmer 2025). But while these involuntary encounters can impose substantial material and social harms on those who directly experience them, most contact with the police is indirect. It is far less clear how the wider community may respond to an increase in stops and arrests, which individuals may read as a justified effort to control crime or as evidence of police overreach. Considerable evidence indicates that the presence of uniformed patrol officers and police vehicles has strong perceptual effects, altering individual behavior and communicating ideas about legitimacy, accountability, and the prevalence of crime (Collazos et al. 2021; Simpson 2017; Telep, Mitchell, and Weisburd 2014). There are several reasons to think that in communities with high crime rates and low levels of trust in law enforcement, sudden changes in the visibility of the police may often have a greater effect on voters than the extent to which they rely on coercion. To motivate this expectation, I draw on three strands of scholarship: reassurance policing (Innes 2004), work on legal cynicism in heavily policed communities (Bell 2017; Carr, Napolitano, and Keating 2007), and research on public perceptions of police conduct (Worden and McLean 2017; Nagin and Telep 2020; National Academies 2018).

First, evidence suggests that encounters with the police, particularly indirect ones, often send ambiguous signals. Subjective evaluations of officer behavior among those who are stopped by the police are only weakly correlated with the ratings of external observers (Worden and McLean 2017), and the evaluations of witnesses themselves diverge greatly, with prior beliefs and contextual information playing a significant

role in shaping whether officers' actions are seen as acceptable and justified (Braga et al. 2014; Nagin and Telep 2020; Waddington et al. 2015). This ambiguity may lead the public to notice changes in the visibility of the police, but be comparatively insensitive to the tactics officers are using. A striking illustration of this comes from an aggressive order-maintenance experiment in Lowell, Massachusetts, where community members noticed that there were more police in the neighborhood, but reported no changes in the demeanor of officers or the strategies they were using (Braga and Bond 2008, 2009).

Second, residents of neighborhoods that are poor or predominantly Black often enter interactions with the police holding deeply ingrained, negative expectations rooted in early socialization and contact with criminal justice institutions (Brunson 2007; Carr, Napolitano, and Keating 2007). These prior expectations may limit the effects of new experiences with the police on attitudes (Anoll and Engelhardt 2023). Indeed, scholars such as Bell argue that the distrust toward the police among poor Black people is connected to a deeper, "marginal and ambivalent relationship with society, the law, and predominant social norms that emanates from institutional and legal failure" (Bell 2017, 2082). In this view, shifts in the behavior of individual officers are unlikely to meaningfully change deeper attitudes that people attach to the police and government institutions.

In line with this, an accumulating experimental record shows that both crackdowns on crime and interventions meant to improve how officers interact with the public often fail to generate substantial or long-lasting effects on evaluations of police fairness or legitimacy (Weisburd et al. 2021; Weisburd et al. 2011; Kochel and Weisburd 2017). A review by the National Academies (2018) echoes this finding, reporting little evidence that proactive police interventions either generate backlash or improve views of law enforcement. Together, this work suggests that the public reacts to sudden changes in the visibility of the police but is less sensitive to short-term changes in the propensity of officers to make stops and arrests.

However, testing these claims poses a variety of challenges. Existing work attempting to measure the character of policing in a given neighborhood often relies on self-reports, which may or may not reflect the actual behavior of officers (Worden and McLean 2017), or infers tactics from administrative data on stops and arrests (e.g. Lerman and Weaver 2014). While large-scale administrative databases offer a more direct measure of police behavior, these data typically are limited to coercive encounters, omitting information on patrol presence and non-enforcement activity. This raises the risk of conflating where officers spend their time with their propensity to stop, search, or arrest. Even benchmark tests designed to recover officer aggression rely on strong, unverifiable assumptions about officer behavior (Neil and Winship 2019). Consistent with these limitations, observational efforts to separate the quality and quantity of policing have yielded mixed results. Lerman and Weaver (2014) find that 311 service requests are associated with both total police stops

and the share involving force or searches, but in opposite directions, while Laniyonu (2019) finds no evidence that the forcefulness of stops predicts lower turnout beyond stop volume itself.

The design of the Jacksonville hotspots experiment addresses many of these limitations by randomly assigning high-crime areas to two interventions that both increased patrol presence while generating sharply different rates of involuntary police encounters. This permits valid estimation of policing's causal effect on voter turnout and an unusually direct comparison of community-based policing to more traditional, enforcement-based tactics. Of course, these treatments are bundled, combining changes to stop and arrest rates, officer presence, personnel selection and training, supervisory directives, and the non-enforcement activities officers performed, and the experiment cannot identify the effect of these components in isolation. Nevertheless, comparing the two arms is informative about whether differences in coercive enforcement, conditional on a substantial increase in visible policing activity, produce meaningfully different participatory effects in the aggregate.

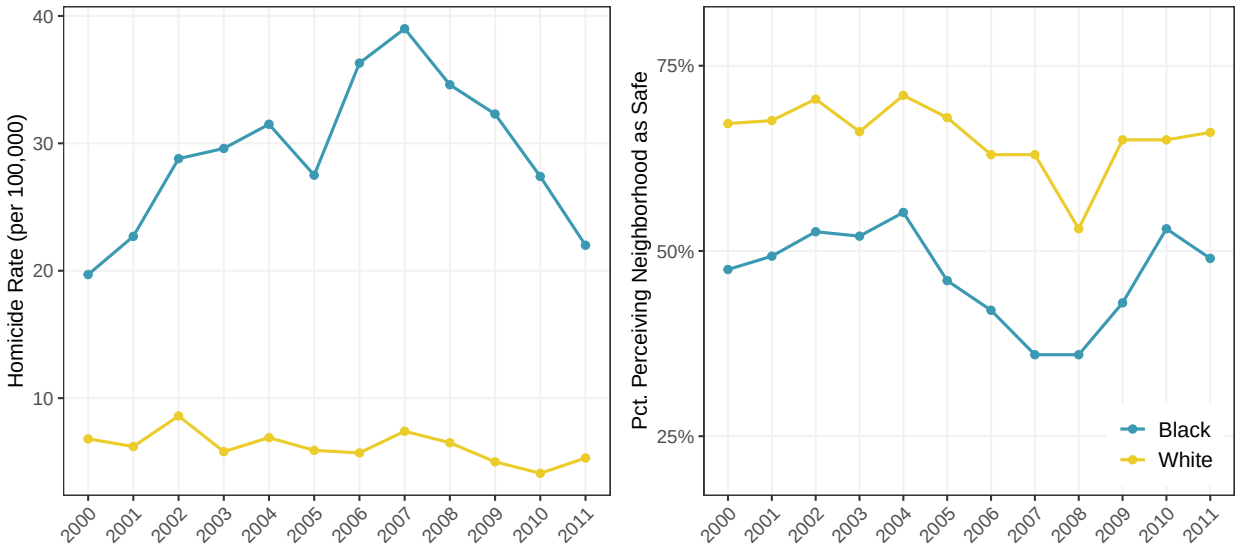
Beyond its methodological appeal, this experiment allows me to measure the impact of a ubiquitous feature of modern American policing. Despite receiving little attention in political science, hotspots-style policing interventions are extremely common — surveys of U.S. law enforcement agencies find that 91% of large agencies report using them to control crime (National Academies 2018). This follows decades of research demonstrating that crime is highly spatially concentrated in small, microgeographic areas — such as specific streets, intersections, and addresses — and that focusing police resources in those places is an efficient and effective way to reduce crime (Braga et al. 2019). These hotspots are highly stable over time and are small enough to suggest that most streets, even in "high-crime" neighborhoods, are relatively free of crime (Weisburd, Groff, and Yang 2012).[1] Scholars also find that attitudes toward the police map closely onto these pockets of crime and disadvantage, often generating more variation in attitudes within neighborhoods than between them (Wheeler et al. 2020; Weisburd et al. 2024). Together, this implies that the micro-geography of crime is a particularly relevant way in which people experience both crime and law enforcement.

Background: The Jacksonville Hotspots Experiment

In the early 2000s, a surge in homicide reversed a decade-long decline in violent crime in Jacksonville, Florida, making public safety a top political concern and earning the city the media moniker "the murder capital of Florida." Mirroring broader patterns of concentrated disadvantage and racial segregation in the city, this violence was heavily concentrated in poor, predominantly Black neighborhoods, exposing Black residents

1. This is often referred to as the "law of crime concentration at places" (Weisburd 2015) given the robustness of the finding.

Figure 1: **Homicide and Perceptions of Safety in Jacksonville, 2000 to 2011**



Note: Panel 1 shows the homicide rate in Duval County by race per 100,000 people. Panel 2 displays the percentage of Blacks and Whites in Duval County who report feeling safe when walking around their neighborhood alone at night.

of Jacksonville to far higher levels of criminal victimization than Whites. As can be seen in Figure 1, the homicide rate for Black people was more than three times greater than for Whites, and Black residents of Jacksonville were far more likely to view their neighborhood as unsafe.[2] In 2007, 63% of Whites in Jacksonville reported feeling safe to walk alone in their neighborhood at night, while only 36% of Blacks did.

In response to public concerns with crime, in 2006 the Duval County Sheriff (who acts as an elected police chief within Jacksonville's combined city-county government) announced an initiative called *Operation Safe Streets* (OSS) which was designed to direct police resources into neighborhoods with high crime rates. The program targeted roughly 55 square miles of the city with a variety of anti-crime measures, including increased patrols, crackdowns on high frequency offenders, gun buy-back programs, and tens of thousands of police door knocks to increase community engagement and crime reporting.[3]

As a way to measure the effectiveness of these various interventions, the Jacksonville Sheriff's Office (JSO) collaborated with external researchers from the Police Executive Research Forum (PERF) to design and conduct a randomized controlled field trial that would compare the relative efficacy of traditional, enforcement-based policing with a more community-based approach to crime reduction. Using Uniform Crime Reports (UCR) data from 2006 to early 2008, analysts within the JSO identified 83 hotspots of "street violence" (i.e. non-domestic violent crime incidents), which averaged 0.02 square miles in size and included a mix of residential

2. Homicide statistics come from the Florida Department of Health's Bureau of Vital Statistics. Survey data comes from the Jacksonville Community Council (JCCI); details given in Appendix I.

3. This original area targeted by OSS was far larger than the hotspots targeted by the experiment.

and nonresidential areas (Taylor, Koper, and Woods 2011).[4] These hotspots were then randomized into one of two policing interventions or a control group for 90 days. Hotspots in the control condition (n = 40) received standard, "business as usual" policing services. Both the existence and location of these sites were known only to a handful of analysts and commanders during the experiment, and subsequent analysis by the PERF researchers found no changes in the number of field stops or self-initiated policing activities in the control hotspots, confirming that routine operations in these areas continued undisturbed (Taylor, Koper, and Woods 2011).

Half of the treated hotspots (n = 21) were assigned to a "saturation" condition, where the presence of officers was increased dramatically, but where officers were not given specific direction on what activities to engage in. Pairs of officers (who were either on-duty or working overtime) were assigned to patrol these areas at "high risk days and times" when crime incidents were most likely to occur, with each location receiving an average of 53 hours of police patrols per week (Taylor, Koper, and Woods 2011). Officers were required to maintain implementation logs, and were tracked by supervisors to ensure that they remained within their assigned hotspots. Adherence to treatment protocols was additionally verified by the PERF researchers through ride-alongs and detailed interviews with officers participating in the study and those in the surrounding patrol beats (Taylor, Koper, and Woods 2011). Consistent with an increase in police presence, this intervention led to a large increase in enforcement activity, including an 85% increase in investigatory stops ($p < 0.01$) and a $\approx 200\%$ increase in "self-initiated activities" which includes traffic stops and other investigatory police actions ($p < 0.001$) (166).

The remaining treated hotspots (n = 22) were assigned to a community-oriented, "problem-solving" intervention that increased the presence of officers, but also made substantial changes to the tactics they used. A total of 60 officers were selected to patrol these areas based on their reputations for being "change makers" and having positive relationships with the community.[5] These officers were paired with dedicated crime analysts, and given three days of intensive training in Problem-Oriented Policing (POP) — a proactive style of policing in which officers work to identify the underlying causes of criminal activity in a given context and develop solutions to address them (Cordner and Biebel 2005). Officers were given wide discretion in how they chose to address crime problems in their assigned hotspots, but were discouraged by project supervisors from relying on traditional enforcement whenever possible (Taylor, Koper, and Woods 2011).

This produced sharply different patterns of officer behavior from the saturation arm — self-initiated activities

4. The number of violent incidents over this period ranged from 9 to 163, a high rate considering that some hotspots were as small as a single intersection.

5. Given these criteria, this can be considered a strong intervention that combined both training and officer selection. In contrast, officers in the saturation condition were chosen based on schedule availability and willingness to work overtime.

increased by only 50% ($p < 0.05$) over baseline, and increases in investigatory stops and arrests were small and statistically insignificant (Taylor, Koper, and Woods 2011). Instead of relying primarily on enforcement, officers engaged in a broad menu of crime-prevention projects, which were documented in regular briefings to the lieutenants in charge of the project. Officers most commonly worked with business owners, property managers, and city agencies to improve security (e.g. improving lighting or installing fences and road barriers); addressed nuisances and signs of physical disorder; conducted outreach to gather community input; and engaged in targeted code enforcement. The JSO devoted approximately 2,100 officer-hours per week to this effort, or about ninety-five hours per hotspot (Taylor, Koper, and Woods 2011). Although officers in this condition were not required to remain inside their assigned hotspots for that entire period, many of their crime mitigation strategies — including resident outreach, property-owner interviews, surveillance, code enforcement, and follow-up inspections — required they be on site. As a result, the community-oriented arm's larger number of dedicated officer-hours should not be interpreted as implying a correspondingly larger increase in officers' on-site presence (or resident exposure) relative to the saturation arm.

In short, the experiment led to large increases in the presence of the police, as well as substantial differences in police behavior across treatment arms. While the effects of both interventions on crime were modest (consistent with either no effect or slight decreases across a variety of crime outcomes) (Braga et al. 2019; Taylor, Koper, and Woods 2011), the surge in officer presence would have been noticeable to residents of the hotspots given the small size of these areas and magnitude of the treatments. Studies of other experimental hotspot interventions find that the public notices these types of changes, which can shift evaluations of neighborhood conditions, the police, and local officials, as well as their willingness to cooperate with officers (e.g. Braga and Bond 2008; Bryant, Collins, and Villa 2014; Blattman et al. 2021; Kochel and Weisburd 2017). My analysis contributes to this body of work by assessing the *political* consequences of these interventions.

Design

I measure the effects of the experiment by comparing turnout in the treated and control hotspots in three elections: the August 2010 statewide primary, the November 2010 general election, and the March 2011 Duval County election. This setting allows me to assess the effects of the intervention on both on- and off-cycle elections, including a local election in which the incumbent sheriff (who was ultimately responsible for the intervention) was up for reelection. I use the 2011 local election to assess the impacts of the experiment on vote choice.

The hotspots treated in this experiment experienced very high rates of violence. In an environment where

shootings and assaults are routine occurrences, even a relatively short-lived surge in visible patrols may raise turnout by dampening fear, encouraging a sense of collective efficacy, or signaling government responsiveness. However, as I outline above, these attempts to address under-policing could also fail to change voting habits, or even depress participation if seen as intrusive.

Expectations for the comparison across treatment arms are similarly divided. Accounts that center officers' conduct, including work on the carceral state, street-level bureaucracy, and procedural justice, imply that the saturation and community-oriented interventions should produce different participatory responses. Yet most individuals interact with policing only indirectly — seeing officers, hearing sirens, or learning about encounters vicariously through neighborhood networks (Anoll, Epp, and Israel-Trummel 2022) — and the cues that the public draws from this exposure are not always clear. The account I develop above suggests these "policy-adjacent" individuals may instead notice the added presence of officers while remaining comparatively insensitive to the tactics they use. It is therefore an open question whether we should expect to see meaningful differences in the effects of these two interventions.

Data

For this study I obtained the 2012 Florida voter file, which includes each voter's race, sex, partisan affiliation, date of registration, and date of birth (FDOS 2012).[6] The voter file includes voter history for the 2010 statewide primary, the November 2010 general election, and the March 2011 local election. I restrict my analysis to turnout in these elections. I also obtained precinct-level election returns and GIS data from the Duval County Supervisor of Elections (Duval County, n.d.).

To identify voters residing in the experimental hotspots, I first geocoded voter addresses using the ArcGIS USA Geocoder API, supplementing this with subaddress-level coordinates from the City of Jacksonville's address locator API. Of the 468,352 voters registered in Jacksonville in both 2008 and 2012, over 99.4% were successfully geocoded. I then merged these records with hotspot shapefiles obtained from the Jacksonville Sheriff's Office via a series of public records requests (JSO, n.d.[b]).

I identify 11,518 registered voters whose address lies within one of the 83 crime hotspots identified by the

6. To mitigate issues of post-treatment registration bias, I focus my analysis on voters who were registered to vote before the 2008 election, several months prior to the start of the experiment. Of course, relying on post-treatment voter information raises concerns that selective attrition from the voter file may be skewing the results. Given that I find a positive effect of the two interventions on turnout, the main concern is that the interventions may have led low-propensity voters to move away or be removed from the voter rolls at higher rates. If this were the case, I would expect turnout in elections prior to the experiment to be higher among treated voters — but as I show in the Appendix (Table C.1), the treatment and control groups are balanced across a variety of voter-level characteristics, including past turnout. Estimating placebo effects on pre-treatment turnout in the March 2007 local, August 2008 primary, and November 2008 general elections likewise yields null estimates for both interventions (Appendix Table C.3). There is also no evidence of differential numbers of pre-treatment registrants across arms (Appendix Figure C.1).

JSO.[7] As can be seen in Table 1, these voters are far more likely to be Black and to be registered as Democrats than the broader population of Jacksonville. Voters in these areas are also slightly younger, more likely to be women, and less likely to have voted in the previous election. While these hotspot locations are not representative of the broader city population, they reflect many neighborhoods that are considered highly policed, given the close relationship between crime, segregation, racialized poverty, and the use of preemptive policing tactics (Sampson, Raudenbush, and Earls 1997; Soss and Weaver 2017; Gelman, Fagan, and Kiss 2007).

Table 1: **Descriptive Statistics of Registered Voters Living Within Crime Hotspots**

	Experimental Hotspots	Rest of Jacksonville
Community-Oriented Treatment	36.9%	-
Saturation Treatment	21.5%	-
Voted in 2008	63.5%	77.7%
Black	73.3%	28.8%
Hispanic	3.4%	3.5%
White	19%	62.3%
Democrat	73.7%	44.9%
Republican	11.3%	37.6%
Male	38.6%	44.8%
Age (mean)	40	45
Total	11,518	454,069

The randomization of the treatment implies that voter characteristics should be unrelated to exposure to either policing intervention. I test this assumption with balance tests on a variety of individual-level pre-treatment characteristics, including age, race, sex, partisanship, and prior turnout. Appendix Table C.1 presents means and normalized differences across covariates for each treatment group.[8] The results suggest that randomization created well balanced groups of voters. Moreover, a joint F-test indicates that these pre-treatment covariates do not predict treatment assignment ($p = 0.255$ for the community-oriented group, and 0.392 for the saturation group).

Finally, I obtained computer-aided dispatch (CAD) records from the Jacksonville Sheriff's Office through a public records request (JSO, n.d.[a]). Like most modern police departments, the JSO tracks every 911 and non-emergency call it receives through a CAD system. Each entry logs the call's time and location, the basis of the call, when an officer was dispatched, when the incident was resolved, and the final disposition (e.g.

7. Nineteen hotspots were located within (and named after) large, garden-style apartment complexes that connect to the street network through a small number of entry/exit points. In these cases, I code all registered voters residing in the complex as living within the hotspot. This assignment rule treats the complex as the relevant unit of exposure, rather than relying on within-complex boundaries that could assign neighbors sharing the same courtyard to different exposure conditions, and that are sensitive to mapping and geocoding errors. Assigning voters based on point locations yields similar results.

8. Following Imbens and Rubin (2015), normalized differences are differences between groups divided by the sample standard deviation.

arrest, citation, investigation, verbal warning, unfounded call, etc.). The system also records this information for officer-initiated calls every time an officer contacts dispatchers by radio or mobile computer. Because of this, the CAD system captures a variety of detective and administrative activities that are missing from official crime reports, along with crime incidents that may have not been officially reported. Although officers may not contact dispatch when engaged in low stakes tasks where they do not feel that safety is a concern, CAD data remains the most detailed, minute-by-minute administrative record of where officers are and how they spend their time (Wu and Lum 2017).

As outlined in Appendix J, I obtained incident-level data on 153,045 citizen-initiated calls and 136,823 officer-initiated calls originating from the experimental hotspots from January 2007 to May 2011.[9] In addition to accounting for potential post-treatment changes in crime and police behavior, these data allow me to replicate the original manipulation checks done by the PERF researchers (Taylor, Koper, and Woods 2011). In Appendix Figure J.1 I find that the saturation condition led to an average of 141 additional officer-initiated calls per hotspot (a 166% increase, $p < 0.05$) during the intervention period, and 27 additional stops, citations, and arrests (a 147% increase, $p < 0.05$). Consistent with officers relying less on enforcement and engaging in problem-solving activities that are less likely to be recorded in the CAD data, the community-oriented condition led to an average of 57 additional officer-initiated calls per hotspot (a 67% increase, $p < 0.05$), and only 6 additional stops, citations, and arrests, which is not statistically significant.

Estimation

To estimate the intent-to-treat effect of the two interventions, I compare turnout in the saturation and community-oriented policing hotspots to turnout in the control hotspots. To account for the clustered design and improve the precision of the estimates I rely on regression adjustment, using the estimator proposed by Lin (2013).[10] Specifically I estimate the following model:

$$Vote_{ih} = \beta COP_h + \beta SAT_h + \beta COP_h \times (\mathbf{X}_i - \overline{\mathbf{X}}_i) + \beta SAT_h \times (\mathbf{X}_i - \overline{\mathbf{X}}_i) + \epsilon_{ih} \quad (1)$$

where i indicates an individual voter, h indicates the hotspot they reside in, COP and SAT are the treatments, and $\mathbf{X}$ is a vector of pre-treatment covariates which are centered and fully interacted with the

9. The JSO would only release incident-level CAD data for specific addresses I supplied them with, and provided the data in the form of PDF-based image files that had been manually redacted to exclude calls that are not considered public record, such as those related to sexual assault. I received over 20,000 pages of data in this format. Due to the extremely costly and time-intensive nature of obtaining these records, my data is limited to incidents which occurred at addresses within the hotspots.

10. Adjusting for cluster size (i.e. the count of registered voters within each hotspot) is necessary because the difference-in-means estimator does not give an unbiased estimate of the ATE under cluster-randomized assignment when clusters contain different numbers of treated units (Aronow and Middleton 2013). This potential source of bias can be mitigated by block randomizing on cluster size (Imai, King, and Nall 2009), which was not done by the original PERF research team.

treatment indicators.[11] In addition to cluster size, racial composition (i.e. percent Black), and pretreatment crime, I control for individual-level characteristics plausibly related to voting behavior, including race, age, gender, household size, and pre-treatment turnout in 2006 and 2008. In Appendix Figure D.1 I present estimates with and without controls. The addition of covariates substantially reduces the variance of the estimates by accounting for other individual-level predictors of turnout, while returning nearly identical point estimates. This minimizes concerns that the results are reliant on functional form assumptions or other forms of model dependence.

Finally, because I am testing the effects of two treatments across three elections, I use a Bonferroni correction for these 6 tests, substituting the 95% confidence interval with the more conservative $1 - \frac{\alpha}{6}$, where $\alpha = 0.05$.

Results

In Figure 2, I present the estimated intent-to-treat effects for both the saturation and community-oriented interventions on individual-level turnout. The horizontal lines represent the adjusted 90% and 95% confidence intervals on the difference between treated and control groups. The point estimates are positive across all three elections. I find that the saturation condition increased turnout in the 2010 statewide primary by 2.1 percentage points and by 4.3 percentage points in the 2010 general election ($p < 0.05$). While the point estimate for the 2011 local election is similar in magnitude, it loses statistical significance after adjusting for multiple hypothesis testing, consistent with large increases in turnout ($\approx$ 5 percentage points), as well as null effects.

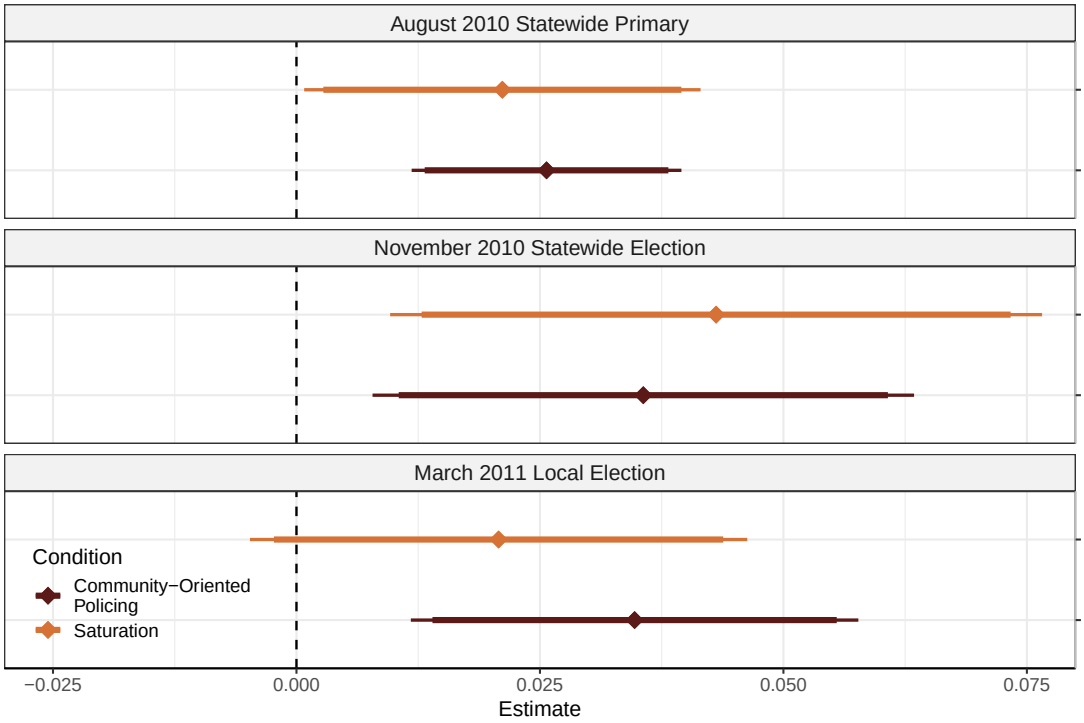
I find that the community-oriented intervention led to similar increases — 2.6 percentage points in the 2010 primary, 3.6 percentage points in the 2010 general, and 3.5 percentage points in the 2011 local election ($p < 0.05$). Given the low turnout in these elections — 22.5% in the 2010 primary and 29.7% in the 2011 election — these are large effects, and surpass the effect sizes of many get-out-the-vote efforts (Green and Gerber 2019). Importantly, the point estimates for both conditions are consistently positive, similar in magnitude, and statistically indistinguishable.

Robustness Checks

I assess the robustness of this finding in several ways. To account for potential spillover effects between the hotspots, I reestimate Equation 1, adding indicators for being within x feet of a treated hotspot. I also

11. This approach is conceptually related to both post-stratification and blocking (Miratrix, Sekhon, and Yu 2013). While my outcome of interest is binary, I rely on OLS, which gives asymptotically correct confidence intervals even under model misspecification (Freedman 2008; Lin 2013).

Figure 2: **Estimated Treatment Effect of Hotspots Policing on Turnout**



Note: Intent-to-treat estimates of the effect of the Jacksonville hotspots policing interventions on voter turnout in elections held in 2010 and 2011 ($n = 11,518$ individuals in 71 crime hotspots). 95% confidence intervals derived from Huber–White heteroscedasticity- and cluster-robust standard errors, along with a Bonferroni correction for 6 hypothesis tests. Full results presented in Appendix Table D.1.

add inverse propensity of treatment weights to account for differences in the probability of being exposed to spillovers, which arise mechanically from the geographic clustering of the intervention sites. I recover the design-based probability of receiving the observed combination of direct and spillover treatments by re-running the original randomization process 5000 times.[12] To avoid extreme weights, I restrict the analysis to hotspots that had between a 10% and 90% probability of receiving the observed combination of treatment and spillovers. I test for spillovers within a 500ft and 1500ft radius, using randomization inference to recover p-values under the sharp null of no effects.[13]

The results are presented in Appendix E. While the trimmed sample leads to a loss of statistical power, I recover very similar estimates of the direct effects of the two treatments when accounting for spillovers. In

12. The original experiment relied on block randomization stratifying on pre-treatment violent street crime counts. While I do not have access to the crime data these strata were constructed from, I create pseudo-blocks using calls for service from January 2007 to May of 2008 related to violence. Any difference between these randomization blocks and the original ones should add noise (making the randomization tests conservative), but not directional bias, as treatment probabilities did not differ across blocks.

13. Randomization inference allows for valid hypothesis testing in this setting by using the experimental design and observed outcomes to generate an empirical distribution of possible treatment effects. Under the sharp null hypothesis of no effect, I re-estimate the weighted models using the 5000 placebo treatment assignments, and measure the percentage of these placebo effects that are at least as large as the observed estimates. Importantly the validity of this approach does not rely on correctly modeling the structure of spillovers (Bowers, Fredrickson, and Panagopoulos 2013).

other words, I find no evidence these mobilizing effects are an artifact of the treatments depressing turnout in nearby control sites, or other forms of spatial interference. Estimated spillover effects are generally small and negative, but inconsistent across elections and treatment arms.

Given the 1 year gap between the conclusion of the experiment and the first election in which I measure turnout, another potential concern is that these effects reflect post-experimental actions taken by the sheriff's department. Because there was some evidence that the community-oriented intervention reduced violent crime, the department attempted to institutionalize this style of policing by setting up a special unit of 20 officers dedicated to problem-solving work (Roush and Koper 2012). A total of 28 hotspots were exposed to this nonexperimental intervention, which could bias my estimates if selection into this intervention was systematically related to treatment assignment. Discussions with former JSO staff indicate that prior to the hotspots experiment, the area covered by Operation Safe Streets had been divided into twelve large "OSS Zones" (see Appendix Figure H.1 for an example). The problem-solving unit focused their efforts in hotspots within four of these zones where crime was thought to be rising. Reassuringly, this indicates that selection into this post-experimental effort was based on geography and neighborhood-level crime rather than treatment assignment or crime conditions in the hotspots themselves.

To verify that this nonrandomized problem-solving initiative is not contaminating my estimates, I estimate the controlled direct effect of the original randomized patrols using the sequential g-estimator (Acharya, Blackwell, and Sen 2016). The estimator models turnout as a function of treatment and exposure to the post-treatment program, subtracts the program's estimated contribution from the outcome, and then re-estimates the treatment effect on this "de-mediated" outcome. As shown in Appendix Table D.6, the treatment coefficients remain virtually unchanged after this adjustment, suggesting that the department's follow-up community-oriented policing program is not driving these effects.[14]

Of course, police behavior could have shifted in other ways after the experiment. Because patrol officers learned which street segments had been classified as "high-crime," they might have focused extra enforcement on those locations once the study ended. Conversely, command staff could have orchestrated crackdowns in the control hotspots, depressing turnout there.[15] I explore this possibility using the JSO's CAD records (JSO, n.d.[a]), focusing on the period from the end of the experiment to the 2010 statewide primary. In the Appendix (Figure J.2), I fail to find clear evidence of systematic treatment-control differences in (1) the number of officer-initiated activities within each hotspot, (2) the share of CAD incidents that are initiated

14. One reason that this later initiative does not appear to have had an impact on voting behavior may be that it was far less intensive than the randomized intervention.

15. There is no documented evidence that this occurred. Moreover, conversations with former JSO staff indicate that there was strong institutional attachment to the much larger OSS Zones used prior to the experiment, and that police work went back to focusing on these areas.

by officers, or (3) the share of officer-initiated activities producing arrests or citations. In short, observable patrol activity after the experiment looks the same in treated and control areas, making it unlikely that unmeasured behavioral changes account for the turnout effects I find.

Heterogeneous Effects

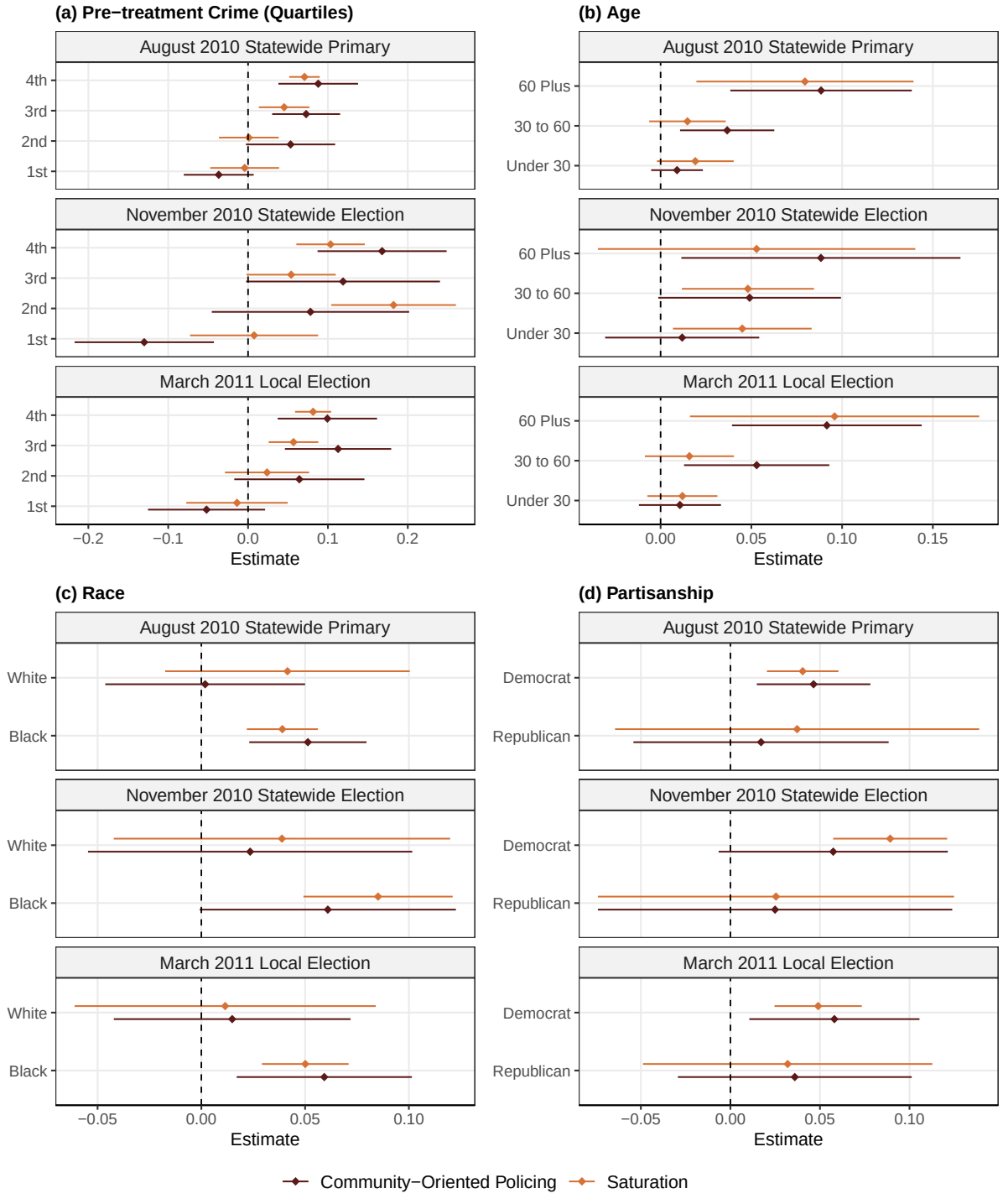
I next turn to heterogeneity in the effect of the interventions. If this increase in turnout was the result of residents feeling safer or better served by law enforcement, I would expect to see the largest turnout increases in hotspots with high levels of pre-treatment crime. Panel A of Figure 3 presents estimated effects of the two treatments by quartiles of pre-treatment, crime-related CAD incidents during the same 90 day window one year before the experiment. In line with this expectation, I find clear evidence of a positive interaction between treatment and crime, with increases in turnout driven by hotspots with higher pre-treatment crime rates.

While I find similar aggregate effects of the two interventions, it is possible that the two interventions mobilized different sets of voters. Across racial groups, older voters consistently report higher satisfaction with law enforcement (Haberman et al. 2016) and are more supportive of increased police patrols and spending (Goldstein 2021; Metcalfe and Bolaji 2024). This is thought to be due in part to the fact that older individuals are more fearful of criminal victimization (Goldstein 2021), and are less likely to be viewed with suspicion by officers (Dunham et al. 2005), to experience involuntary police contact (Davis, Whyde, and Langton 2018), or to report being treated unfairly by the police (Bjornstrom 2015). This suggests older people are both more predisposed to favor high levels of police presence, and less likely to experience the negative consequences of increased enforcement. Given this, we may expect the saturation condition to have been particularly mobilizing for this group.

Panel B of Figure 3 presents heterogeneous effects by age (full results in Appendix Table D.3). Both interventions were more mobilizing for voters over the age of 60. For the community-oriented condition, the interaction between treatment and age is significant in all three elections; for the saturation condition, it is significant in the 2010 primary and the 2011 local election, but not the 2010 general. In Panels C and D I further explore potential sources of heterogeneity by race and partisanship. While the small number of Republican and non-Black voters in the sample yields imprecise estimates for these groups, the results show clear evidence of mobilization among Black voters across both treatment arms. Importantly, I fail to find evidence indicating that the two interventions were systematically mobilizing different sets of voters.

Finally, in the Appendix (Table F.1), I examine whether the experiment had differential effects outside the

Figure 3: **Heterogeneous Effects**



Note: OLS estimates with 95% confidence intervals derived from Huber–White heteroscedasticity- and cluster-robust standard errors. Full results presented in Appendix Tables D.2, D.3, D.4, and D.5.

hotspots. In their original analysis, the PERF researchers found some evidence of an increase in violence-related calls for service within 500ft of the hotspots, which could be the result of crime displacement or a greater willingness to report crime (Taylor, Koper, and Woods 2011). As with the interference tests presented above, I use a combination of inverse propensity of treatment weights and randomization inference to estimate spillover effects among registered voters outside the hotspot boundaries within a 500ft or 1500ft radius, comparing them to individuals within the same distance of a control site. I find no statistically significant effects on turnout in the surrounding areas, with estimates close to zero. In short, there is no evidence that the interventions had substantial impacts on the surrounding community.

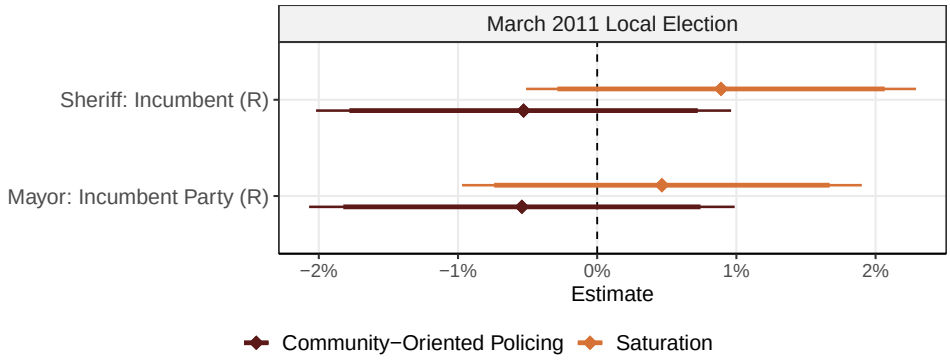
Vote Choice

I next turn to the effects of the interventions on vote choice. A large literature shows that voters often engage in retrospective evaluations of basic government services, as well as social and economic conditions, and reward the officials they believe are responsible for improvements (e.g. Burnett and Kogan 2017). If residents viewed their neighborhood as safer following the intervention, this may have translated into electoral rewards for incumbents, especially in local elections given that policing policy is set at the local level and public safety is often an important issue in local races (Brown and Zoorob 2022). The 2011 election may have been particularly conducive to attribution given that Republicans held unified control over city government at the time, and the incumbent Republican sheriff, who serves as the head of the Jacksonville police, was on the ballot.

Because the hotspots are far smaller than election precincts, I use the geocoded voter file to calculate the percentage of voters registered before the 2008 election within each precinct that live within an experimental hotspot. I also calculate the percentage of voters that live within these hotspots that were assigned to either the community-oriented or saturation condition and take this as my main treatment measure.[16] I additionally control for the percentage of voters who were registered as Democrats in each precinct before the start of the experiment, and merge the precinct boundaries with 2010 Census data (US Census Bureau 2010) to account for racial composition (i.e. percent Black), the density of children, and concentrated disadvantage (i.e. rates of home vacancy, single-parent households, unemployment, college attainment, and welfare recipiency). The results, presented in Figure 4, show that coverage by either condition is not associated with voting for the incumbent sheriff, or the incumbent mayoral party (full results in Appendix Table G.1). These null results are robust to dropping both partisan and demographic controls, as well as calculating hotspot coverage in terms of geographic area (i.e. square miles).

16. This mechanically limits the analysis to the 63 precincts (24.6% of total) that contain at least one registered voter living in a crime hotspot.

Figure 4: **Exposure to Treatment and Support for Local Incumbents**



Note: OLS estimates and 90% and 95% confidence intervals of the association between treatment coverage and precinct-level vote choice in the March 2011 Duval County election. Treatment coverage is standardized so that estimates represent the change in vote share associated with a one standard deviation increase in the share of registered voters in violent crime hotspots assigned to each experimental condition. Full results presented in Appendix Table G.1.

Alternate Mechanisms

Could the observed gains in participation reflect backlash rather than reassurance? Previous work finds that sudden surges in police activity can sometimes backfire by *increasing* fear of crime (Hinkle and Weisburd 2008), which in turn may mobilize those concerned with disorder (Brown and Zoorob 2022). To test this possibility I again turn to the CAD data, examining citizen-initiated calls for "minor," nuisance-related issues. If residents had become more fearful, I would expect more of these calls in the treated areas relative to calls about more serious crimes, which suffer from fewer issues of under-reporting. I rely on several measures of call inflation, including the proportion of citizen-initiated calls rated as low priority by dispatchers,[17] the ratio of calls about violent incidents relative to those about suspicious-persons or vehicles, the proportion of calls determined to be unverified or unfounded, and the proportion of calls diverted to the department's Teleserv unit, which handles certain low priority complaints by phone, rather than dispatching officers. I find null effects of the two treatments on all four outcomes (Appendix Figure J.3). Each of these measures is imperfect — for example, the rate of unverified calls is likely the product of the true crime rate, the propensity of citizens to call the police, and officer response times. However, the consistency of this null result across measures gives little reason to think the patrol surge heightened fear. More broadly, qualitative evidence suggests that residents of the crime hotspots were both aware of, and concerned with, the violence occurring in their communities. Media interviews of residents of the Grand Park neighborhood — the location of one of the control hotspots — document how residents feared spending time outside in the neighborhood due

17. While there may be racial and gender bias in the types of incidents that dispatchers consider to be serious, this would only pose a threat to inference if this bias varied systematically with treatment assignment. Because dispatchers were not directly involved with the experiment, this seems unlikely.

to the violence, and expressed that they would feel safer if the police spent more time in the neighborhood rather than reacting to crimes after they occurred (Schoettler 2010). This was a frequent complaint among residents of neighborhoods with high crime rates, with one complaining that "[i]t takes a kid to get shot to bring the police here...[t]hey only show up when something happens" (Lewis 2006). In this environment, it is highly unlikely that increasing police presence would inflate perceptions of crime. Indeed, existing experimental evidence in criminology indicates these "backfire" effects are uncommon and largely limited to places with far lower baseline crime rates (e.g. Weisburd et al. 2021).

Another potential explanation is that this mobilization is the result of backlash to excessive police surveillance, particularly in the saturation condition which generated large numbers of stops and arrests. Existing scholarship finds that mobilization against police violence and overreach is usually driven by young, infrequent voters and accompanied by other signs of civic resistance such as protests (Ang and Tebes 2024; Naftel, *Forthcoming*). As demonstrated in Figure 3, I fail to find similar demographic turnout patterns here. To further investigate potential backlash I draw on data from Olzak (2021) on protests against police violence in major American cities from 1990 to 2018, which come from manually coded news reports from *Newsbank*. Because this data is based on a random sample of years and focuses specifically on violence, I supplement it with additional news reports between 2006 and 2011 from four Jacksonville newspapers: *The Florida Times-Union*, *The Jacksonville Free Press*, *The Florida Star*, and the *Jacksonville Daily Record*.[18] Following Olzak (2021), I searched digitized archives for each paper using various string combinations (e.g. "demonstrat* AND police," "protest* AND police," "police brutality," "police-involved shooting") to identify potential protest activity related to police brutality, discrimination, or overreach.[19] While newspapers often fail to cover small, nonviolent demonstrations, relying on multiple local papers which vary in their editorial stance, topical focus, and audience demographics reduces the likelihood of false negatives (Earl et al. 2004).

I find no documented police-related protest activity in the years leading up to the experiment, or in the post-treatment period from 2009 to 2011 — indeed, several articles remarked on the conspicuous *lack* of protests against police violence during this period given the city's high rate of fatal police shootings (Galnor 2008). While it is possible that some individuals were mobilized by these concerns, media accounts from the time suggest that residents responded positively to the increase in police patrols. Interviews in the summer of 2009 with residents of an apartment complex treated by the saturation intervention discuss how the sudden influx of officers discouraged loitering and made residents feel safe enough to walk to the local park. One resident remarked, "[p]olice made their presence felt, and now the people are taking back their home"

18. Both *The Jacksonville Free Press* and *The Florida Star* are historically Black newspapers.
19. News reports were accessed via the following sources: ProQuest, Florida Times-Union Archives (Newsbank), and the Florida Newspaper Archive.

(Coleman 2009).[20] In an article on the Eastside neighborhood — the location of one of the community-oriented policing hotspots — a member of the local neighborhood association credited the neighborhood's decrease in crime to the police, who had become more visible, closed a well-known drug house, and increased their community engagement (Patton 2009).

A final possibility is that the results reflect a monitoring effect, due to officers behaving more respectfully and professionally while under closer observation during the experiment. While greater oversight can alter how officers engage with the public (Mummolo 2018), the monitoring in the saturation condition was meant to ensure adherence to the assigned patrol schedules, not to alter officers' demeanor or their stopping rates. A 2012 agency-wide survey found that 80% of officers believed supervisors routinely used technology to track their daily activities (Koper et al. 2015), suggesting the experiment should not have materially changed oversight. And while it's possible the experiment led to a short-term decline in more serious acts of misconduct, these behaviors are likely too rare for these changes to have been noticeable within areas as small as the intervention sites.[21]

Discussion and Conclusion

This article provides the first field-experimental evidence that police deployments can alter the composition of the electorate. While a large body of work documents the demobilizing power of police encounters (e.g. Ben-Menachem and Morris 2023; Lerman and Weaver 2020), far less is known about how the state's attempts to remedy under-policing might affect political behavior. Leveraging the randomized design of the Jacksonville Hotspots Experiment (Taylor, Koper, and Woods 2011) allows me to bring experimental evidence to bear on these questions, and to compare the political effects of two very different approaches to crime control. I demonstrate that in neighborhoods experiencing high levels of violence, large increases in police patrols produced durable boosts in turnout, particularly among Black voters. Although voting is only one form of civic engagement, it is particularly consequential in the low-salience, off-cycle elections examined here, where the electorate is disproportionately affluent and White (Hajnal and Trounstine 2005). Strikingly, this mobilization occurred regardless of whether officers relied on traditional patrols or community-oriented tactics.

The similar mobilizing effects of the two interventions are particularly surprising given that the saturation

20. There was no indication those living in the hotspots were aware that this sudden increase in policing activity was connected to an experiment. A search through local newspapers uncovered only two brief articles that mentioned the experiment specifically — both referred to the intervention as a community policing initiative and neither noted it was a part of a randomized controlled trial.

21. Internal affairs data from the JSO show the department investigated roughly 600 officer misconduct complaints per year in the three years preceding the intervention. Even if the true rate of misconduct was twice as high, the probability of witnessing one of these incidents within a given hotspot during the 90 day experiment would have been vanishingly small.

arm generated four times as many stops, citations, and arrests as the community-oriented intervention — a stark difference in enforcement driven by officer selection, training, and supervisory directives. Of course, this finding does not imply that the saturation condition did not generate harms, particularly among those who directly experienced these involuntary encounters. Individuals most at risk of involuntary police contact often have prior experiences with the criminal justice system (Brunson 2007), and are less likely to be registered to vote. If the saturation arm demobilized the individuals who were stopped and arrested, this effect could be understated, or missed entirely, in an analysis restricted to registered voters. This possibility cautions against treating communities experiencing concentrated disadvantage and violence as homogeneous places, and underscores the need for theory and empirical designs that help determine the conditions under which specific groups are (de)mobilized by law enforcement.

At the same time, the absence of an aggregate turnout penalty is an important finding. Both interventions increased turnout, and did so in similar ways, primarily mobilizing older and Black registrants as well as those in the highest-crime hotspots. This convergent pattern of mobilization is difficult to reconcile with accounts in which the two experimental arms mobilized voters via different mechanisms. In other words, it appears unlikely that voters were reacting specifically to the punitive elements of the saturation arm and the problem-solving approaches of the community-oriented alternative.[22] One plausible reading is that residents of the hotspots reacted primarily to elements common to both treatments, such as increased police visibility or other salient signals of state attention and responsiveness. This interpretation aligns with recent work in policy feedback theory showing that citizens' reactions to government action are powerfully conditioned by context and preexisting attitudes, often in ways that diverge from what we would expect based on formal policy design features alone (Anzia, Jares, and Malhotra 2022), as well as with classic accounts of symbolic politics that argue that visible government actions help drive what citizens expect out of government by signaling threat or reassurance (Edelman 1985).

Drawing on insights from work on procedural justice (Nagin and Telep 2020) and legal estrangement (Bell 2017), I argue that attempts to change how officers interact with the public may be difficult for the "policy adjacent" to observe and relatively easy to discount, given strongly held beliefs about law enforcement that are rooted in history and accumulated experiences. In communities that feel abandoned by law enforcement, the visible presence of the police may send a far stronger signal. My findings are consistent with this argument

22. An alternate explanation is that the similar turnout effects across arms reflect offsetting forces (e.g., the saturation arm was less mobilizing due to the coercive contact it generated, but was also more visible). This would have required the officers in the saturation arm to have spent meaningfully more public-facing time in the hotspots than those in the community-oriented arm, and that additional presence to have had a non-trivial marginal effect at patrol levels that were already very high. While the two experimental arms were not designed to be exactly matched on patrol hours, dedicated officer hours were, if anything, greater in the community-oriented condition. These offsetting forces would also have needed to be roughly stable across elections, places, and voter subgroups to yield the similar aggregate and heterogeneous turnout patterns I observe.

and enrich theoretical accounts of indirect contact with the police, while prompting deeper questions about what people are reacting to when we observe empirical links between policing and political behavior. Policing is, of course, a bundled treatment, and while the setting studied here offers unusually detailed measures of where officers were and the activities they engaged in, I am unable to test whether these responses were purely due to an increase in police visibility, or to other components of these interventions. Regardless of the interpretation of the results, they suggest that community-level responses to policing are not as sensitive to police tactics as is often assumed, and that these responses are not reducible to the involuntary encounters experienced by others in the community. Those not at high personal risk of being stopped by the police may respond more strongly to perceived changes in the provision of basic security functions than to proximity to coercion, even in communities with strained relationships with law enforcement.

While I identify a meaningful increase in turnout as a result of these interventions, no single experiment can definitively establish *why* sudden surges in police activity produced this effect. The evidence I present weighs against a simple backlash story, and the lack of change in the long-term crime rate after the experiment makes it unlikely that these effects are a direct consequence of lower rates of victimization. There is, however, suggestive evidence pointing toward a set of positive feedback mechanisms, such as increased political efficacy, reduced fear of victimization, or improved perceptions of government responsiveness. Prior studies show that visible police patrols are viewed favorably by most residents of crime hotspots, which can boost their regard for the police and their sense of collective efficacy (Koper et al. 2022; Kochel and Weisburd 2019). Contemporaneous news coverage from Jacksonville points in the same direction: residents described safer streets, fewer loiterers, and a general feeling that order had improved as a result of the deployment. Reductions in visible disorder are not well captured by official crime statistics, but can weigh heavily on people's sense of security (Skogan 1992), which may help explain how the deployments shifted political behavior without producing measurable declines in crime. It is also possible that the surges themselves served as a catalyzing event, spurring individuals to organize around neighborhood problems and engage local government. Disentangling these mechanisms provides a promising path for future research.

Another important limitation is the intervention's relatively short 90 day duration. A longer-lasting deployment might have amplified the mobilization effects documented here, especially if sustained exposure led to deeper changes in crime patterns, community dynamics, and views of law enforcement. Conversely, an extended police saturation-style intervention may have generated backlash by leading residents to eventually view high-intensity patrols as an occupation. Nevertheless, the fact that turnout effects endure nearly two years later suggests that even relatively brief interventions may recalibrate residents' political engagement

in a lasting way.[23]

In principle, the approach used here — linking experimental hotspots to geocoded administrative microdata — could be applied to study other downstream outcomes where comparable data exist. Replication of this analysis in other settings could also help clarify scope conditions, though doing so presents several challenges. Despite the large number of hotspot experiments conducted over the past two decades, many are poorly suited to this style of analysis, featuring short-lived interventions, modest sample sizes, or crossover designs in which the same locations alternate between treatment and control, complicating the construction of a clean comparison group for outcomes like voting that may be measured years after the intervention period. Even when an experiment is substantively well suited to this approach, reconstruction can be difficult. Geographic and treatment-assignment data needed for record linkage are often not retained; when they are, they are rarely public and typically held by police departments whose cooperation can be hard to secure. Realizing the broader promise of this approach will require stronger data-preservation and replication standards in criminology.

Readers may wonder if these results would generalize to more politically progressive cities or to the present day in the wake of the *Black Lives Matter* movement. In the Appendix, I demonstrate that police killings were a salient local issue at the time of the experiment, and that perceptions of police discrimination among Black residents of Jacksonville were significantly higher than in the nation as a whole (Appendix Figure I.1). This suggests the findings are not merely an artifact of a uniquely pro-police environment; rather, the mobilizing potential of law enforcement likely depends on baseline levels of (in)security. We should expect very different reactions to the presence of the police in neighborhoods with low crime rates, or in cases where officers resort to extreme, racialized tactics that place entire communities at risk of harm (Lerman and Weaver 2014; Naftel, *Forthcoming*).

Finally, these findings leave open important questions about electoral attribution. Despite a sizable increase in turnout, I detect no effects of the treatments on vote choice. This may be due to the difficulty voters have in correctly attributing performance to local government officials (de Benedictis-Kessner 2018), or a product of the particularly strong racial and partisan polarization in Jacksonville. However, the absence of an electoral reward for an intervention that successfully expanded participation is itself an important finding. It suggests an accountability asymmetry that merits greater attention in future work on urban governance.

This asymmetry matters for how governments choose among competing approaches to policing high-crime neighborhoods. The surplus stops and arrests generated by the saturation arm do not appear to have

23. This is consistent with previous work that finds both extreme (Ang and Tebes 2024) and low-level (Ben-Menachem and Morris 2023) police encounters can generate participatory effects that persist through multiple election cycles.

generated any additional reductions in long-run crime or increases in turnout relative to the community-oriented arm. If, as I suggest, voters responded primarily to visible signals of state attention, investments beyond policing could potentially generate similar participatory gains while avoiding some of the harms associated with involuntary contact. However, these types of investments may often require more time, money, and institutional coordination than simply putting additional officers on the street. Given this, local officials may have incentives to pursue policies that are comparatively simple and inexpensive to implement, but that concentrate burdens on communities with the least electoral influence.

These findings arrive amid an intense national debate about over- and under-policing in the United States. Proponents of "defunding" the police stress the social costs of aggressive enforcement, while opponents warn that withdrawing officers will leave communities unprotected from crime. The results reported here complicate these claims. Surges in patrols increased civic participation, but only in the neighborhoods suffering the highest levels of violent crime, and they did so regardless of the tactics officers used. This does not imply that police tactics never matter, but it does suggest that public reactions to law enforcement are powerfully shaped by context. In communities subjected to violence and neglect, simply seeing uniformed officers may send signals about government capacity and responsiveness that can mobilize voters.

By demonstrating that coercive institutions can, under specific conditions, bring marginalized citizens back into the electoral process, the study invites scholars to rethink how public safety, inequality, and the carceral state intersect in the urban United States and beyond.

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The author declares no ethical issues or conflicts of interest in this research.

Ethical Standards

The author affirms this research did not involve human participants.

Data Availability Statement

Research documentation and data that support the findings of this study are openly available at the American Political Science Review Dataverse: https://doi.org/10.7910/DVN/9QV9DL.

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Supplementary Material

Policing, Protection, and Participation:

The Effects of a Randomized 'Hotspots' Experiment on Voter Turnout

Daniel Naftel

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B Existing Evidence on the Political Feedback Effects of Policing

Table B.1 summarizes existing empirical scholarship on the relationship between policing and political behavior (e.g. voter turnout, protesting) or political attitudes (e.g. trust in government, political efficacy, support for social movements). By "policing" I mean an action taken by a police agency or officer in the field. This includes police stops, patrols, deployments, hiring (as a measure of street-level presence), raids, crackdowns, curfew enforcement, killings, beatings, and uses of force, but excludes downstream consequences of these encounters such as incarceration. I further exclude studies whose only outcome is calls for service (311 or 911 calls) as these are considered a poor proxy for civic engagement (White and Trump 2018).

Table B.1: **Previous Studies on Policing and Political Behavior**

Citation	Variation	Location	Design	Outcome
Police violence / lethal force				
Williamson et al. (2018)	Police killings (muni. rate)	U.S.	XC	↑ Protesting
Burch (2022)	Police killings (nbhd. count)	U.S.	Natural exp.	↑ Turnout; ↑ political interest (Black respondents)[a]
Silva et al. (2022)	Police stop (nbhd. rate)	U.S.	XC	↓ Trust in govt.
Branton et al. (2023)	Police killings (nbhd. rate)	U.S.	XC	↓ Perceived efficacy
Markarian (2023)	Police killings (proximity)	U.S. (CA, FL, OH)	DiD	↓ Turnout
Ang & Tebes (2024)	Police killings (proximity)	Los Angeles	DiD	↑ Turnout; ↑ registration
de Kadt et al. (2024)	Police massacre (proximity)	South Africa	DiD	↓ Incumb. party vote share; ↑ new party vote share
Drolc & Shoub (2024)	Police killings (proximity)	Los Angeles + U.S.	DiD + survey exp.	↓ Trust in govt.
Morris & Shoub (2024)	Police killings (proximity)	U.S.	RDiT	↑ Turnout
Reid (2024)	Police killings (county rate)	U.S.	XC	↕ Vote for Dem. cand.[b]
Branton & Carey (2025)	Police killings (county rate)	U.S.	XC	↑ Turnout; ↑ non-voting part.
Hassel & Holbein (2026)	Police killing (George Floyd murder)	U.S.	RDiT	↑ Registration
Police repression / crackdowns				
Aytaç et al. (2018)	Police crackdown	Istanbul, Turkey	Survey exp.	↑ Protest willingness
Muñoz & Anduiza (2019)	Police violence (national event)	Spain	Natural exp.	↑ Movement support
Curtice & Behlendorf (2021)	Police violence	Uganda	Survey exp.	↑ Protest willingness
Rodon & Guinjoan (2022)	Police raids (muni., binary)	Catalonia, Spain	XC	↕ Turnout[c]

Continued on next page

Table B.1: **Previous Studies on Policing and Political Behavior** (Continued)

Citation	Variation	Location	Design	Outcome
Aksoy et al. (2023)	Curfew enforcement (muni., binary)	Turkey	DiD + panel	↕ Turnout; ↕ incumb. party vote intention[d]
Naftel (*Forthcoming*)	Anti-gang crackdowns (within-nbhd., binary)	Los Angeles	DiD	↑ Turnout; ↑ registration; ↑ non-voting part.
Individual police stops (direct, involuntary contact)				
Weaver & Lerman (2010)	Police stop (self rep.)	U.S.	XC + matching	↓ Turnout; ↓ trust in govt.
Drakulich et al. (2017)	Police stop (self rep.)	U.S.	XC	↑ Vote intention; ↑ political interest
Anoll et al. (2022)	Police stop (self rep.)	U.S. (IL, NC)	XC	↑ Turnout; ↑ non-voting part.
Christiani & Shoub (2022)	Police stop (self rep.)	U.S.	XC + survey exp.	↑ Non-voting part.
Silva et al. (2022)	Police stop (self rep.)	U.S.	XC	↕ Trust in govt.[e]
Ben-Menachem & Morris (2023)	Traffic stop (admin.)	Hillsborough, FL	DiD	↓ Turnout
Turner & Shum (2025)	Police stop (self rep.)	South Bend, IN	XC	↑ Non-voting part.; ∅ trust in govt.
Best (2026)	Police contact (self rep.)	U.S.	XC	↑ Non-voting part. (Black women)
Police presence, enforcement, and deployments				
Kang & Dawes (2017)	Police stops (nbhd. rate)	New York City	Panel	↓ Turnout
Blair et al. (2019)	Police patrols (community deployment)	Liberia	RCT	∅ Trust in govt.
Condra et al. (2019)	Police presence (nbhd. deployment)	Afghanistan	Quasi-exp.	↓ Turnout
Laniyonu (2019)	Police stops (nbhd. rate)	New York City	XC + matching	↕ Turnout; ↕ mayoral vote share[f]
Anoll et al. (2022)	Police stops (muni. racial disp.)	U.S. (IL, NC)	XC	∅ Non-voting part.
Niven (2022)	Police presence (polling place)	Homewood, AL	XC	↓ Turnout (Black voters)
Turner & Shum (2025)	Police presence (station prox.)	South Bend, IN	XC	↑ Non-voting part.; ↑ trust in govt.
English et al. (2025)	Heavy police presence (during protest)	U.S.	Survey exp.	∅ Protest support
McCarthy et al. (2025)	Police stops (nbhd. rate)	Chicago	XC	↓ Turnout
Palmer (2025)	Police stops (nbhd. rate)	New York City	Panel	↓ Turnout
Romero (2025)	Police force size (muni. count)	U.S.	IV-DiD	↑ Turnout (Black voters)

Designs: XC = cross-sectional; DiD = difference-in-differences; RDiT = regression discontinuity in time; IV = instrumental variables; IV-DiD = instrumented difference-in-differences
Outcomes: ↑ = positive effect or association (more participation/engagement/trust); ↓ = negative; ∅ = null; ↕ = direction varies across elections, subgroups, or moderating conditions

Notes:

[a] Burch (2022) finds a statistically significant increase in turnout and political interest among Black respondents under 40 living near officer-involved deaths of low-threat Black victims that trended on Google.

[b] Reid (2024) finds a statistically significant relationship in 2020, but not 2016.

[c] Rodon & Guinjoan (2022) find police raids on polling stations reduced turnout in directly affected municipalities but raised turnout in nearby ones.

[d] Aksoy et al. (2023) find both outcomes varied by ethnic alignment — curfews depressed turnout among Kurdish (out-group) voters, who disengaged from politics, while raising support for the governing AKP among co-ethnic (in-group) voters.

[e] Silva et al. (2022) find that being questioned on foot was associated with higher trust in local government, whereas vehicle stops were associated with lower trust among Black respondents.

[f] Laniyonu (2019) finds stop-and-frisk intensity predicted lower turnout in off-year national elections (2006, 2010) but higher turnout in 2008 and in the 2013 municipal races where candidates campaigned on SQF reform; it also predicted greater support for the anti-SQF mayoral candidate and less support for the pro-SQF candidate in that election.

C Balance Checks

In their original analysis of the Jacksonville Hotspots Experiment, Taylor, Koper, and Woods (2011) fail to find substantial pre-treatment differences between the three treatment arms across a wide range of measures of crime and officer behavior. This includes self-initiated policing activity, police field stops, calls for service, arrests, and UCR crime incidents (162). Hotspots in the three conditions were also similar in terms of their geographic size and physical characteristics (i.e. residential, commercial, or mixed-use).

In Table C.1 I compare the average pre-treatment characteristics of individual voters using the normalized differences. The normalized difference is given by $\Delta_{norm} = (\overline{X_c} - \overline{X_t})/(\sqrt{(s_c^2 + s_t^2)/2}\,)$ where $\overline{X}$ is the group mean and s is the sample standard deviation (Imbens and Rubin 2015). While any form of imbalance across experimental conditions can induce estimation error (Imai, King, and Stuart 2008; Miratrix, Sekhon, and Yu 2013), I assess whether these imbalances are sufficiently large to induce model dependence and pose potential threats to inference using the rule of thumb suggested by Imbens and Rubin (2015) of 0.25. I find no imbalances in individual-level characteristics such as race, gender, age, partisan affiliation, and turnout history that meet this threshold.

Table C.1: **Covariate Balance Across Experimental Conditions**

	Control	COP	Saturation	Control-COP	Control-Saturation	COP-Saturation
Variables	Mean			Normalized Diff.		
Age	38.62	40.51	41.05	[-0.114]	[-0.150]	[-0.032]
Race:Black	0.71	0.78	0.70	[-0.160]	[0.013]	[0.173]
Male	0.35	0.40	0.40	[-0.103]	[-0.106]	[-0.003]
Democrat	0.72	0.77	0.72	[-0.113]	[-0.012]	[0.101]
Republican	0.12	0.10	0.12	[0.073]	[0.021]	[-0.052]
Voted (2008)	0.65	0.62	0.65	[0.063]	[0.003]	[-0.060]
Voted (2006)	0.19	0.23	0.21	[-0.084]	[-0.055]	[0.029]
N. Voters	344.31	363.61	347.78	[-0.080]	[-0.014]	[0.089]

Figure C.1: **Distribution of Cluster Size by Treatment Assignment**

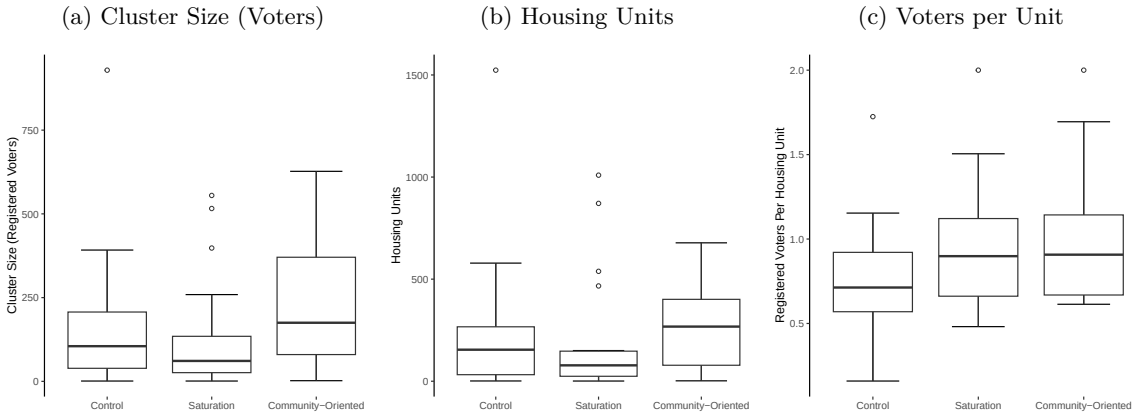


Table C.2 presents regression-based balance tests, predicting treatment assignment with the full vector of individual- and cluster-level covariates. No covariates significantly predict treatment assignment at the 0.05 level and the results of an F-test of joint significance suggest that the sample is balanced along these observed characteristics.

Lastly, in Figure C.1 I compare the distribution of cluster size across the three experimental arms, along with cluster-level counts of housing units, and registered voters per housing unit. The results suggest that there are not any extreme outliers with high leverage that could induce model dependence.

Table C.2: **Tests for Pre-Treatment Balance**

	COP	Saturation
(Intercept)	-0.28 [-1.37, 0.81]	0.30 [-0.46, 1.06]
Voted (2008)	-0.05 [-0.11, 0.02]	0.01 [-0.03, 0.05]
Voted (2006)	0.04 [-0.01, 0.09]	-0.01 [-0.04, 0.02]
Democrat	0.03 [-0.01, 0.06]	-0.01 [-0.03, 0.02]
Republican	0.01 [-0.06, 0.08]	-0.02 [-0.06, 0.02]
Age	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
Male	0.03 [-0.03, 0.10]	0.02 [-0.01, 0.05]
Race:Black	0.09 [-0.12, 0.30]	-0.03 [-0.14, 0.07]
Cluster Size (logged)	0.09 [-0.13, 0.31]	-0.02 [-0.18, 0.15]
Adj. R2	0.04	0.00
N. Observations	11365	11365
N. Clusters	71	71
F-test p-value	0.255	0.392

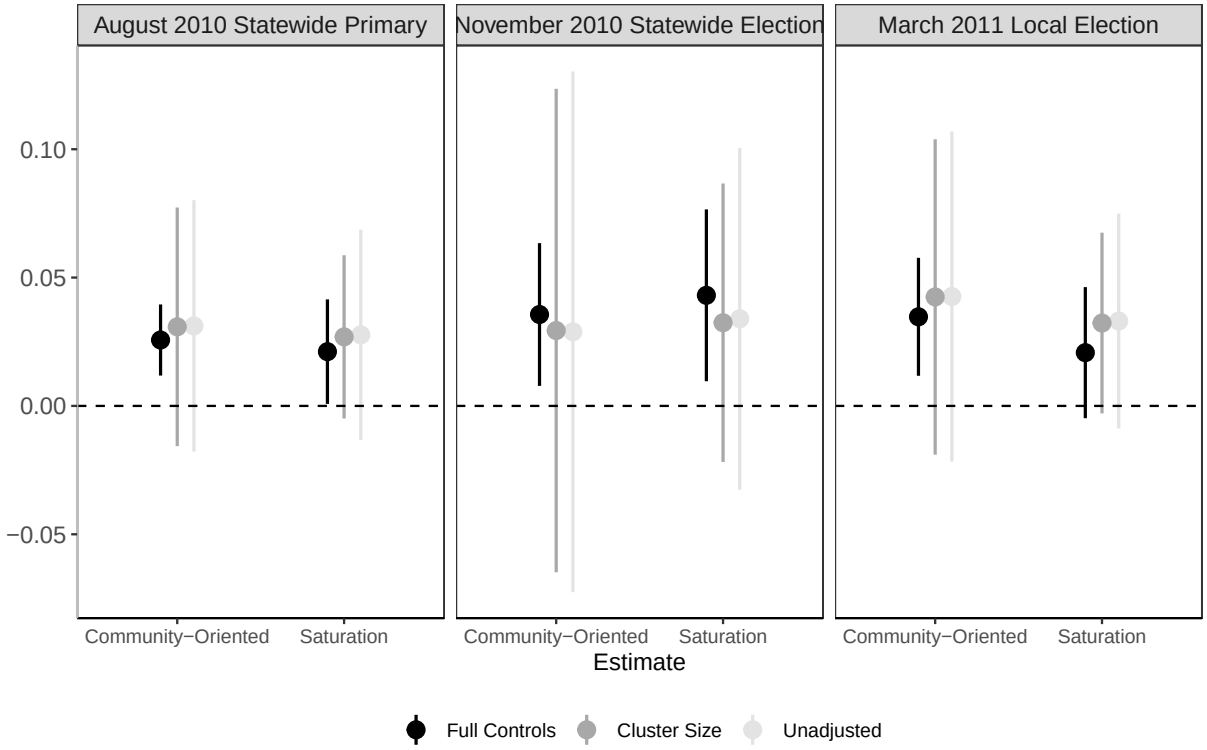
Table C.3: **Placebo Treatment Effects of Hotspots Policing on Pre-Treatment Turnout**

	March 2007 Local Election	August 2008 Primary Election	November 2008 General Election
Intercept	**0.021** [0.010, 0.031]	**0.077** [0.066, 0.089]	**0.630** [0.589, 0.671]
COP	0.003 [-0.010, 0.016]	0.012 [-0.004, 0.028]	-0.019 [-0.082, 0.043]
SAT	0.004 [-0.010, 0.018]	0.009 [-0.008, 0.026]	0.017 [-0.030, 0.064]
Male	0.001 [-0.010, 0.012]	-0.007 [-0.021, 0.007]	**-0.077** [-0.118, -0.037]
Democrat	0.004 [-0.019, 0.027]	**0.029** [0.003, 0.055]	**0.088** [0.037, 0.140]
Republican	0.003 [-0.019, 0.025]	0.029 [-0.011, 0.070]	**0.101** [0.017, 0.186]
Age	**0.001** [0.000, 0.002]	**0.002** [0.001, 0.003]	0.001 [-0.001, 0.003]
Race: Black	-0.015 [-0.032, 0.002]	**-0.039** [-0.072, -0.005]	**0.151** [0.096, 0.207]
Voted (2006)	**0.076** [0.049, 0.103]	**0.198** [0.154, 0.241]	**0.303** [0.236, 0.369]
HH Size: 2	0.001 [-0.019, 0.021]	0.015 [-0.012, 0.042]	**0.061** [0.008, 0.114]
HH Size: 3–4	-0.010 [-0.034, 0.013]	0.005 [-0.031, 0.042]	0.051 [-0.014, 0.116]
HH Size: 5+	-0.022 [-0.056, 0.013]	-0.052 [-0.117, 0.013]	-0.015 [-0.169, 0.139]
HH Size: Missing	-0.007 [-0.026, 0.011]	**-0.037** [-0.066, -0.009]	-0.034 [-0.207, 0.139]
Cluster Size	-0.000 [-0.000, 0.000]	0.000 [-0.000, 0.000]	0.000 [-0.000, 0.000]
Prop. Black (Cluster)	0.010 [-0.030, 0.050]	0.053 [-0.016, 0.122]	-0.199 [-0.433, 0.036]
Crime Calls (2008)	0.000 [-0.000, 0.000]	0.000 [-0.000, 0.000]	-0.000 [-0.000, 0.000]
Male × COP	-0.009 [-0.024, 0.006]	-0.006 [-0.034, 0.021]	0.029 [-0.023, 0.081]
Democrat × COP	0.000 [-0.028, 0.028]	-0.011 [-0.066, 0.045]	0.011 [-0.090, 0.112]
Republican × COP	-0.003 [-0.039, 0.032]	-0.028 [-0.091, 0.034]	-0.034 [-0.174, 0.105]
Age × COP	0.000 [-0.001, 0.001]	0.000 [-0.001, 0.002]	-0.001 [-0.004, 0.002]
Race: Black × COP	0.016 [-0.006, 0.039]	0.039 [-0.014, 0.093]	-0.008 [-0.108, 0.092]
Voted (2006) × COP	-0.002 [-0.042, 0.037]	0.059 [-0.014, 0.132]	0.052 [-0.040, 0.144]
HH Size: 2 × COP	0.010 [-0.021, 0.040]	0.008 [-0.048, 0.064]	-0.032 [-0.105, 0.040]
HH Size: 3–4 × COP	-0.013 [-0.043, 0.017]	0.005 [-0.047, 0.057]	-0.035 [-0.116, 0.046]
HH Size: 5+ × COP	-0.001 [-0.051, 0.049]	0.031 [-0.052, 0.113]	0.013 [-0.177, 0.202]
HH Size: Missing × COP	0.004 [-0.026, 0.033]	0.039 [-0.014, 0.093]	-0.088 [-0.320, 0.144]
Cluster Size × COP	0.000 [-0.000, 0.000]	-0.000 [-0.000, 0.000]	-0.000 [-0.001, 0.000]
Prop. Black (Cluster) × COP	0.000 [-0.047, 0.048]	**-0.118** [-0.198, -0.038]	-0.029 [-0.310, 0.252]
Crime Calls (2008) × COP	-0.000 [-0.000, 0.000]	-0.000 [-0.000, 0.000]	**0.000** [0.000, 0.001]
Male × SAT	-0.011 [-0.028, 0.006]	-0.020 [-0.058, 0.019]	0.029 [-0.039, 0.098]
Democrat × SAT	-0.006 [-0.037, 0.026]	-0.027 [-0.064, 0.011]	0.035 [-0.058, 0.127]
Republican × SAT	-0.010 [-0.054, 0.033]	0.011 [-0.060, 0.083]	0.004 [-0.109, 0.116]
Age × SAT	-0.000 [-0.001, 0.001]	0.000 [-0.001, 0.002]	-0.000 [-0.002, 0.002]
Race: Black × SAT	0.013 [-0.018, 0.044]	**0.055** [0.006, 0.104]	-0.003 [-0.109, 0.103]
Voted (2006) × SAT	-0.000 [-0.032, 0.031]	**0.074** [0.003, 0.145]	0.001 [-0.077, 0.079]
HH Size: 2 × SAT	-0.005 [-0.039, 0.030]	-0.011 [-0.071, 0.048]	-0.046 [-0.116, 0.025]
HH Size: 3–4 × SAT	0.009 [-0.029, 0.046]	-0.032 [-0.099, 0.035]	-0.059 [-0.187, 0.069]
HH Size: 5+ × SAT	0.023 [-0.068, 0.115]	0.051 [-0.031, 0.134]	-0.019 [-0.238, 0.201]
HH Size: Missing × SAT	0.066 [-0.069, 0.202]	**0.197** [0.016, 0.377]	0.020 [-0.181, 0.221]
Cluster Size × SAT	0.000 [-0.000, 0.000]	-0.000 [-0.000, 0.000]	0.000 [-0.000, 0.000]
Prop. Black (Cluster) × SAT	0.005 [-0.060, 0.071]	-0.029 [-0.175, 0.118]	0.021 [-0.292, 0.335]
Crime Calls (2008) × SAT	-0.000 [-0.000, 0.000]	0.000 [-0.000, 0.000]	-0.000 [-0.001, 0.001]
Adj. R^2	0.068	0.185	0.117
N. Voters	9,149	10,373	11,282
N. Hotspots	71	71	71

Note: OLS estimates with 95% confidence intervals derived from Huber–White heteroscedasticity- and cluster-robust standard errors, with a Bonferroni adjustment for six hypothesis tests. Estimates in bold have adjusted confidence intervals that exclude zero.

D Main Results

Figure D.1: **Intent-to-Treat Effects of Experiment**



Note: Intent-to-treat estimates of the effect of the Jacksonville hotspots policing interventions on voter turnout in the 2010 and 2011 elections ($n = 11,365$ individuals in 71 crime hotspots). Covariate-adjusted estimates are reported, as well as estimates from models that include either limited (i.e. cluster size only) or no pre-treatment controls. 95% confidence intervals derived from Huber–White heteroscedasticity- and cluster-robust standard errors, with a Bonferroni adjustment for six hypothesis tests

Table D.1: **Estimated Treatment Effect of Hotspots Policing on Turnout**

	2010 Primary Election	2010 General Election	2011 Local Election
Intercept	**0.085** [0.075, 0.095]	**0.243** [0.224, 0.263]	**0.126** [0.111, 0.140]
COP	**0.026** [0.012, 0.040]	**0.036** [0.008, 0.063]	**0.035** [0.012, 0.058]
SAT	**0.021** [0.001, 0.041]	**0.043** [0.010, 0.077]	0.021 [-0.005, 0.046]
Male	-0.003 [-0.020, 0.013]	**-0.032** [-0.064, -0.001]	-0.011 [-0.038, 0.017]
Democrat	**0.036** [0.012, 0.059]	0.022 [-0.004, 0.049]	**0.029** [0.001, 0.058]
Republican	0.060 [-0.001, 0.120]	0.053 [-0.015, 0.121]	0.017 [-0.012, 0.047]
Age	**0.003** [0.002, 0.003]	**0.004** [0.003, 0.005]	**0.002** [0.001, 0.003]
Race: Black	-0.021 [-0.052, 0.011]	-0.004 [-0.031, 0.022]	-0.015 [-0.042, 0.012]
Voted (2006)	**0.208** [0.158, 0.258]	**0.365** [0.303, 0.427]	**0.286** [0.242, 0.329]
Voted (2008)	**0.065** [0.047, 0.083]	**0.225** [0.186, 0.264]	**0.099** [0.078, 0.120]
HH Size: 2	0.007 [-0.023, 0.038]	0.037 [-0.004, 0.078]	0.007 [-0.032, 0.046]
HH Size: 3–4	0.008 [-0.032, 0.047]	**0.054** [0.004, 0.104]	0.014 [-0.028, 0.056]
HH Size: 5+	-0.025 [-0.139, 0.089]	0.030 [-0.094, 0.154]	-0.019 [-0.135, 0.097]
HH Size: Missing	-0.033 [-0.071, 0.006]	**-0.105** [-0.137, -0.073]	**-0.068** [-0.105, -0.030]
Cluster Size	-0.000 [-0.000, 0.000]	-0.000 [-0.000, 0.000]	-0.000 [-0.000, 0.000]
Prop. Black (Cluster)	0.024 [-0.048, 0.097]	-0.069 [-0.170, 0.033]	0.042 [-0.045, 0.129]
Crime Calls (2008)	-0.000 [-0.000, 0.000]	-0.000 [-0.000, 0.000]	-0.000 [-0.000, 0.000]
Male × COP	0.010 [-0.020, 0.039]	0.040 [-0.002, 0.083]	0.017 [-0.015, 0.049]
Democrat × COP	-0.004 [-0.041, 0.033]	0.001 [-0.049, 0.051]	0.020 [-0.027, 0.068]
Republican × COP	0.000 [-0.098, 0.098]	0.008 [-0.082, 0.098]	0.028 [-0.030, 0.086]
Age × COP	0.001 [-0.000, 0.002]	0.000 [-0.001, 0.002]	0.000 [-0.001, 0.002]
Race: Black × COP	0.044 [-0.002, 0.090]	0.035 [-0.011, 0.081]	0.032 [-0.009, 0.073]
Voted (2006) × COP	0.051 [-0.025, 0.127]	0.020 [-0.080, 0.121]	0.038 [-0.052, 0.128]
Voted (2008) × COP	0.004 [-0.033, 0.040]	0.016 [-0.062, 0.093]	0.022 [-0.023, 0.067]
HH Size: 2 × COP	0.012 [-0.037, 0.062]	-0.020 [-0.077, 0.037]	0.041 [-0.013, 0.094]
HH Size: 3–4 × COP	-0.001 [-0.055, 0.053]	0.009 [-0.063, 0.081]	0.036 [-0.032, 0.105]
HH Size: 5+ × COP	0.032 [-0.094, 0.159]	0.029 [-0.113, 0.172]	0.104 [-0.041, 0.249]
HH Size: Missing × COP	0.038 [-0.025, 0.100]	0.045 [-0.020, 0.111]	0.051 [-0.013, 0.114]
Cluster Size × COP	0.000 [-0.000, 0.000]	0.000 [-0.000, 0.000]	0.000 [-0.000, 0.000]
Prop. Black (Cluster) × COP	-0.070 [-0.154, 0.014]	0.012 [-0.111, 0.135]	-0.059 [-0.162, 0.043]
Crime Calls (2008) × COP	-0.000 [-0.000, 0.000]	0.000 [-0.000, 0.000]	-0.000 [-0.000, 0.000]
Male × SAT	-0.013 [-0.040, 0.015]	0.020 [-0.021, 0.061]	0.009 [-0.049, 0.066]
Democrat × SAT	-0.018 [-0.048, 0.013]	0.033 [-0.017, 0.084]	0.006 [-0.044, 0.056]
Republican × SAT	-0.002 [-0.130, 0.126]	0.011 [-0.079, 0.100]	0.027 [-0.054, 0.109]
Age × SAT	0.001 [-0.001, 0.002]	0.000 [-0.002, 0.002]	**0.001** [0.000, 0.003]
Race: Black × SAT	0.007 [-0.034, 0.048]	0.017 [-0.039, 0.073]	0.033 [-0.025, 0.092]
Voted (2006) × SAT	0.031 [-0.059, 0.121]	-0.049 [-0.153, 0.055]	-0.030 [-0.130, 0.070]
Voted (2008) × SAT	0.015 [-0.021, 0.050]	0.015 [-0.069, 0.099]	0.024 [-0.030, 0.077]
HH Size: 2 × SAT	0.005 [-0.056, 0.067]	-0.024 [-0.080, 0.032]	0.008 [-0.061, 0.078]
HH Size: 3–4 × SAT	0.012 [-0.061, 0.084]	-0.016 [-0.103, 0.070]	0.014 [-0.065, 0.093]
HH Size: 5+ × SAT	0.013 [-0.111, 0.136]	0.065 [-0.151, 0.281]	0.106 [-0.030, 0.241]
HH Size: Missing × SAT	0.121 [-0.037, 0.278]	0.095 [-0.049, 0.239]	0.081 [-0.112, 0.273]
Cluster Size × SAT	-0.000 [-0.000, 0.000]	0.000 [-0.000, 0.000]	-0.000 [-0.000, 0.000]
Prop. Black (Cluster) × SAT	0.084 [-0.072, 0.240]	0.071 [-0.154, 0.296]	0.014 [-0.129, 0.158]
Crime Calls (2008) × SAT	0.000 [-0.000, 0.000]	-0.000 [-0.001, 0.000]	0.000 [-0.000, 0.001]
Adj. R^2	0.219	0.311	0.236
N. Voters	11,365	11,365	11,365
N. Hotspots	71	71	71

Note: OLS estimates with 95% confidence intervals derived from Huber–White heteroscedasticity- and cluster-robust standard errors, with a Bonferroni adjustment for six hypothesis tests. Estimates in bold have adjusted confidence intervals that exclude zero.

Table D.2: **Heterogeneous Effects by Pre-treatment Crime**

	2010 Primary Election	2010 General Election	2011 Local Election
Intercept	**0.105** [0.087, 0.123]	**0.308** [0.282, 0.335]	**0.154** [0.126, 0.181]
COP	-0.037 [-0.081, 0.008]	**-0.130** [-0.219, -0.041]	-0.052 [-0.127, 0.023]
Cluster Size	-0.000 [-0.000, 0.000]	**-0.000** [-0.000, -0.000]	-0.000 [-0.000, 0.000]
Prop. Black (Cluster)	-0.008 [-0.079, 0.064]	-0.073 [-0.197, 0.051]	0.024 [-0.055, 0.104]
Crime: 2nd Quartile	-0.029 [-0.059, 0.002]	**-0.079** [-0.121, -0.036]	-0.037 [-0.075, 0.000]
Crime: 3rd Quartile	**-0.036** [-0.065, -0.007]	**-0.086** [-0.147, -0.024]	**-0.052** [-0.086, -0.019]
Crime: 4th Quartile	**-0.061** [-0.086, -0.035]	**-0.153** [-0.197, -0.108]	**-0.064** [-0.097, -0.032]
SAT	-0.004 [-0.048, 0.040]	0.008 [-0.074, 0.089]	-0.014 [-0.078, 0.051]
Cluster Size × COP	0.000 [-0.000, 0.000]	-0.000 [-0.000, 0.000]	-0.000 [-0.000, 0.000]
Prop. Black (Cluster) × COP	-0.086 [-0.198, 0.027]	-0.075 [-0.289, 0.140]	-0.094 [-0.242, 0.054]
Crime: 2nd Quartile × COP	**0.090** [0.021, 0.159]	**0.208** [0.058, 0.358]	**0.116** [0.010, 0.223]
Crime: 3rd Quartile × COP	**0.109** [0.046, 0.173]	**0.249** [0.094, 0.404]	**0.165** [0.062, 0.267]
Crime: 4th Quartile × COP	**0.125** [0.052, 0.197]	**0.298** [0.167, 0.429]	**0.151** [0.044, 0.258]
Cluster Size × SAT	**-0.000** [-0.000, -0.000]	-0.000 [-0.000, 0.000]	**-0.000** [-0.000, -0.000]
Prop. Black (Cluster) × SAT	**0.109** [0.013, 0.205]	0.087 [-0.095, 0.269]	0.079 [-0.083, 0.242]
Crime: 2nd Quartile × SAT	0.005 [-0.043, 0.053]	**0.175** [0.089, 0.261]	0.038 [-0.031, 0.106]
Crime: 3rd Quartile × SAT	0.049 [-0.003, 0.101]	0.046 [-0.058, 0.151]	**0.071** [0.001, 0.141]
Crime: 4th Quartile × SAT	**0.075** [0.019, 0.131]	0.096 [-0.012, 0.204]	**0.095** [0.013, 0.178]
Adj. R^2	0.010	0.017	0.010
N. Voters	11,518	11,518	11,518
N. Hotspots	71	71	71

Note: OLS estimates with 95% confidence intervals derived from Huber–White heteroscedasticity- and cluster-robust standard errors. Estimates in bold have confidence intervals that exclude zero. The reference category is the 1st quartile of pre-treatment crime.

Table D.3: **Heterogeneous Effects by Age**

	2010 Primary Election	2010 General Election	2011 Local Election
Intercept	**0.013** [0.004, 0.022]	**0.102** [0.078, 0.127]	**0.037** [0.027, 0.048]
COP	0.009 [-0.005, 0.024]	0.012 [-0.031, 0.055]	0.011 [-0.012, 0.034]
Cluster Size	0.000 [-0.000, 0.000]	0.000 [-0.000, 0.000]	0.000 [-0.000, 0.000]
Prop. Black (Cluster)	-0.008 [-0.057, 0.040]	-0.081 [-0.203, 0.041]	0.021 [-0.059, 0.102]
Crime Calls (2008)	-0.000 [-0.000, 0.000]	**-0.000** [-0.000, -0.000]	-0.000 [-0.000, 0.000]
Age: 30–60	**0.076** [0.057, 0.095]	**0.175** [0.143, 0.208]	**0.104** [0.085, 0.122]
Age: 60+	**0.231** [0.197, 0.265]	**0.360** [0.300, 0.420]	**0.245** [0.199, 0.291]
SAT	0.019 [-0.002, 0.041]	**0.045** [0.006, 0.084]	0.012 [-0.008, 0.032]
Cluster Size × COP	0.000 [-0.000, 0.000]	-0.000 [-0.000, 0.000]	-0.000 [-0.000, 0.000]
Prop. Black (Cluster) × COP	-0.064 [-0.135, 0.008]	-0.035 [-0.214, 0.143]	-0.065 [-0.186, 0.057]
Crime Calls (2008) × COP	0.000 [-0.000, 0.000]	**0.000** [0.000, 0.001]	0.000 [-0.000, 0.000]
Age: 30–60 × COP	0.028 [-0.002, 0.057]	0.037 [-0.017, 0.091]	**0.042** [0.006, 0.079]
Age: 60+ × COP	**0.079** [0.028, 0.131]	**0.076** [0.002, 0.151]	**0.081** [0.026, 0.136]
Cluster Size × SAT	-0.000 [-0.000, 0.000]	-0.000 [-0.000, 0.000]	-0.000 [-0.000, 0.000]
Prop. Black (Cluster) × SAT	0.066 [-0.033, 0.164]	0.100 [-0.093, 0.292]	0.043 [-0.083, 0.169]
Crime Calls (2008) × SAT	0.000 [-0.000, 0.000]	-0.000 [-0.001, 0.000]	**0.000** [0.000, 0.001]
Age: 30–60 × SAT	-0.004 [-0.028, 0.019]	0.003 [-0.035, 0.041]	0.004 [-0.024, 0.031]
Age: 60+ × SAT	**0.060** [0.001, 0.120]	0.008 [-0.084, 0.099]	**0.084** [0.003, 0.165]
Adj. R^2	0.088	0.095	0.077
N. Voters	11,518	11,518	11,518
N. Hotspots	71	71	71

Note: OLS estimates with 95% confidence intervals derived from Huber–White heteroscedasticity- and cluster-robust standard errors. Estimates in bold have confidence intervals that exclude zero. The reference category is registered voters under the age of 30.

Table D.4: **Heterogeneous Effects by Race**

	2010 Primary Election	2010 General Election	2011 Local Election
Intercept	**0.128** [0.101, 0.155]	**0.278** [0.240, 0.316]	**0.159** [0.134, 0.185]
COP	0.002 [-0.047, 0.051]	0.024 [-0.056, 0.103]	0.015 [-0.043, 0.073]
Cluster Size	-0.000 [-0.000, 0.000]	0.000 [-0.000, 0.000]	0.000 [-0.000, 0.000]
Prop. Black (Cluster)	0.032 [-0.039, 0.102]	-0.068 [-0.189, 0.054]	0.042 [-0.041, 0.126]
Crime Calls (2008)	**-0.000** [-0.000, -0.000]	**-0.000** [-0.001, -0.000]	**-0.000** [-0.000, -0.000]
Race: Black	**-0.069** [-0.096, -0.042]	**-0.066** [-0.098, -0.034]	**-0.057** [-0.085, -0.030]
Race: Hispanic	**-0.074** [-0.115, -0.033]	**-0.136** [-0.203, -0.069]	**-0.107** [-0.132, -0.082]
Race: Other	**-0.094** [-0.122, -0.066]	**-0.115** [-0.163, -0.068]	**-0.085** [-0.117, -0.052]
SAT	0.042 [-0.018, 0.101]	0.039 [-0.044, 0.121]	0.011 [-0.062, 0.085]
Cluster Size × COP	-0.000 [-0.000, 0.000]	-0.000 [-0.001, 0.000]	-0.000 [-0.000, 0.000]
Prop. Black (Cluster) × COP	**-0.130** [-0.242, -0.018]	-0.099 [-0.313, 0.115]	-0.128 [-0.282, 0.027]
Crime Calls (2008) × COP	**0.000** [0.000, 0.000]	**0.001** [0.000, 0.001]	**0.000** [0.000, 0.001]
Race: Black × COP	**0.049** [0.000, 0.099]	0.037 [-0.040, 0.114]	0.044 [-0.013, 0.102]
Race: Hispanic × COP	-0.029 [-0.131, 0.074]	-0.091 [-0.211, 0.030]	-0.013 [-0.103, 0.077]
Race: Other × COP	0.005 [-0.044, 0.053]	-0.048 [-0.130, 0.035]	-0.024 [-0.093, 0.044]
Cluster Size × SAT	**-0.000** [-0.000, -0.000]	-0.000 [-0.000, 0.000]	**-0.000** [-0.000, -0.000]
Prop. Black (Cluster) × SAT	**0.113** [0.018, 0.209]	0.117 [-0.082, 0.317]	0.060 [-0.084, 0.204]
Crime Calls (2008) × SAT	**0.000** [0.000, 0.001]	0.000 [-0.000, 0.001]	**0.000** [0.000, 0.001]
Race: Black × SAT	-0.002 [-0.055, 0.050]	0.046 [-0.031, 0.124]	0.039 [-0.029, 0.106]
Race: Hispanic × SAT	**-0.064** [-0.126, -0.002]	-0.018 [-0.109, 0.073]	-0.020 [-0.079, 0.038]
Race: Other × SAT	0.003 [-0.053, 0.060]	-0.018 [-0.104, 0.067]	0.023 [-0.061, 0.106]
Adj. R^2	0.017	0.022	0.014
N. Voters	11,518	11,518	11,518
N. Hotspots	71	71	71

Note: OLS estimates with 95% confidence intervals derived from Huber–White heteroscedasticity- and cluster-robust standard errors. Estimates in bold have confidence intervals that exclude zero. The reference category for race is White.

Table D.5: **Heterogeneous Effects by Partisanship**

	2010 Primary Election	2010 General Election	2011 Local Election
Intercept	0.010 [-0.004, 0.024]	**0.129** [0.095, 0.163]	**0.050** [0.032, 0.069]
COP	0.012 [-0.007, 0.031]	0.013 [-0.040, 0.066]	-0.005 [-0.028, 0.019]
Cluster Size	-0.000 [-0.000, 0.000]	0.000 [-0.000, 0.000]	0.000 [-0.000, 0.000]
Prop. Black (Cluster)	-0.016 [-0.081, 0.050]	-0.109 [-0.233, 0.016]	-0.001 [-0.085, 0.082]
Crime Calls (2008)	**-0.000** [-0.000, -0.000]	**-0.000** [-0.001, -0.000]	**-0.000** [-0.000, -0.000]
Democrat	**0.062** [0.048, 0.076]	**0.096** [0.068, 0.124]	**0.065** [0.046, 0.085]
Republican	**0.125** [0.076, 0.175]	**0.166** [0.103, 0.229]	**0.094** [0.070, 0.118]
SAT	**0.032** [0.009, 0.055]	0.022 [-0.036, 0.079]	0.005 [-0.028, 0.039]
Cluster Size × COP	-0.000 [-0.000, 0.000]	-0.000 [-0.001, 0.000]	-0.000 [-0.000, 0.000]
Prop. Black (Cluster) × COP	-0.097 [-0.212, 0.018]	-0.081 [-0.323, 0.160]	-0.108 [-0.276, 0.060]
Crime Calls (2008) × COP	**0.000** [0.000, 0.000]	**0.001** [0.000, 0.001]	**0.000** [0.000, 0.001]
Democrat × COP	0.034 [-0.004, 0.073]	0.044 [-0.011, 0.099]	**0.063** [0.012, 0.114]
Republican × COP	0.005 [-0.076, 0.086]	0.012 [-0.085, 0.108]	0.041 [-0.018, 0.099]
Cluster Size × SAT	**-0.000** [-0.000, -0.000]	-0.000 [-0.000, 0.000]	**-0.000** [-0.000, -0.000]
Prop. Black (Cluster) × SAT	**0.111** [0.009, 0.213]	0.123 [-0.076, 0.321]	0.079 [-0.066, 0.224]
Crime Calls (2008) × SAT	**0.000** [0.000, 0.001]	0.000 [-0.000, 0.001]	**0.001** [0.000, 0.001]
Democrat × SAT	0.008 [-0.020, 0.036]	**0.068** [0.019, 0.116]	**0.044** [0.006, 0.082]
Republican × SAT	0.005 [-0.098, 0.109]	0.004 [-0.077, 0.085]	0.027 [-0.051, 0.105]
Adj. R^2	0.023	0.028	0.020
N. Voters	11,518	11,518	11,518
N. Hotspots	71	71	71

Note: OLS estimates with 95% confidence intervals derived from Huber–White heteroscedasticity- and cluster-robust standard errors. Estimates in bold have confidence intervals that exclude zero. The reference category is voters not registered with either major political party.

Table D.6: **Average Treatment Effects and Controlled Direct Effects**

Election	Condition	ATE	ACDE
2010 Primary	Community-Oriented	**0.026** [0.012, 0.040]	**0.024** [0.002, 0.059]
2010 Primary	Saturation	**0.021** [0.001, 0.041]	0.022 [-0.002, 0.061]
2010 General	Community-Oriented	**0.036** [0.008, 0.063]	0.038 [-0.017, 0.068]
2010 General	Saturation	**0.043** [0.010, 0.077]	0.042 [-0.023, 0.081]
2011 Local	Community-Oriented	**0.035** [0.012, 0.058]	**0.036** [0.005, 0.077]
2011 Local	Saturation	0.021 [-0.005, 0.046]	0.020 [-0.019, 0.053]

Note: For ATE: 95% confidence intervals derived from Huber–White heteroscedasticity- and cluster-robust standard errors, with a Bonferroni adjustment for six hypothesis tests. For ACDE: 95% cluster-bootstrap confidence intervals. Estimates in bold have confidence intervals that exclude zero.

E Spatial Interference

Table E.1: **Spillovers at 500ft**

	2010 Primary Election		2010 General Election		2011 Local Election		
	Estimate	RI p-value	Estimate	RI p-value	Estimate	RI p-value	N. Hotspots
Direct Effects:							
Saturation	**0.030**	0.037	**0.045**	0.035	0.025	0.138	11
Community-Oriented	0.024	0.076	0.030	0.142	0.030	0.075	15
Indirect Effects:							
Saturation	0.006	0.664	-0.035	0.087	0.019	0.239	3
Community-Oriented	0.002	0.871	-0.027	0.163	-0.012	0.487	6
						Tot. Hotspots:	59

Table E.2: **Spillovers at 1500ft**

	2010 Primary Election		2010 General Election		2011 Local Election		
	Estimate	RI p-value	Estimate	RI p-value	Estimate	RI p-value	N. Hotspots
Direct Effects:							
Saturation	0.029	0.065	**0.043**	0.048	0.025	0.156	8
Community-Oriented	0.030	0.051	0.026	0.211	0.031	0.074	10
Indirect Effects:							
Saturation	-0.004	0.659	**-0.037**	0.021	0.001	0.919	8
Community-Oriented	-0.006	0.564	-0.002	0.886	**-0.026**	0.032	11
						Tot. Hotspots:	50

Note: OLS estimates weighted by the inverse probability of observed joint exposure to treatment and spillovers. The sample is trimmed to hotspots with exposure probabilities between 0.10 and 0.90. Two-sided randomization-inference p-values are based on 5,000 simulations of the original randomization procedure. Estimates in bold indicate that $p \leq 0.05$. Negative spillovers at 1,500 ft are statistically significant in one of three elections for both treatment arms, but are not detected among nearby voters outside the experimental hotspots (Table F.1). This finding may reflect sampling variability in sparsely populated exposure cells.

F Neighborhood Spillovers

Table F.1: **Neighborhood Spillovers**

	2010 Primary Election		2010 General Election		2011 Local Election	
	Estimate	RI p-value	Estimate	RI p-value	Estimate	RI p-value
Within 500ft:						
Saturation	0.011	0.291	0.001	0.903	-0.005	0.673
Community-Oriented	-0.001	0.958	-0.010	0.383	-0.015	0.207
Within 1500ft:						
Saturation	0.006	0.296	0.003	0.620	0.001	0.857
Community-Oriented	0.006	0.369	0.008	0.146	-0.007	0.342

Note: OLS estimates weighted by the inverse probability of observed joint exposure to treatment and spillovers. The sample is trimmed to voters with exposure probabilities between 0.10 and 0.90 (n = 11,460/48,047). Two-sided randomization-inference p-values are based on 5,000 simulations of the original randomization procedure. Estimates in bold indicate that $p \leq 0.05$.

G Vote Choice

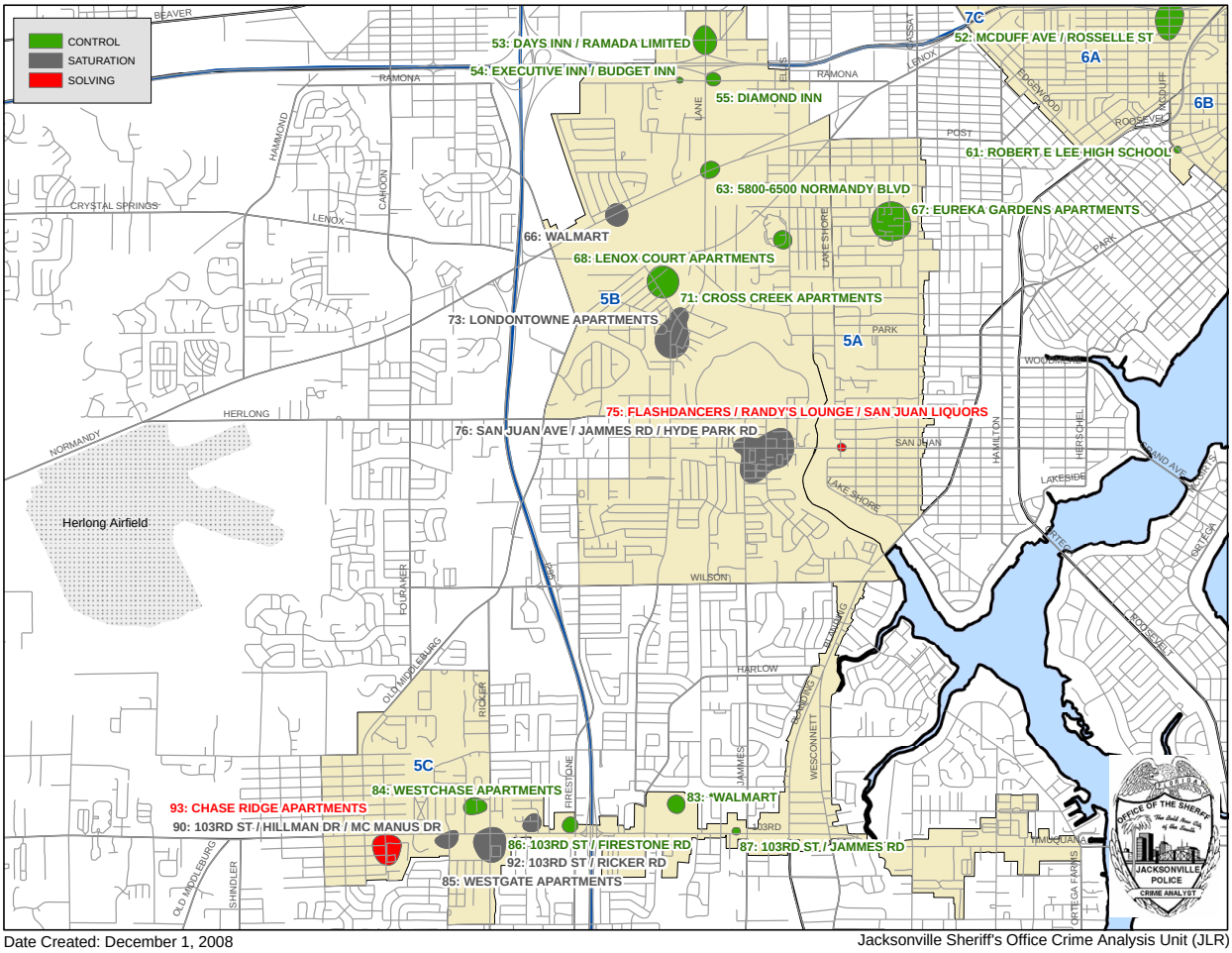
Table G.1: **Vote Choice (Precinct-Level)**

	Sheriff	Mayor
Intercept	**1.276** [1.109, 1.444]	**1.327** [1.155, 1.499]
Community-Oriented Share (1 SD)	-0.005 [-0.020, 0.010]	-0.005 [-0.021, 0.010]
Saturation Share (1 SD)	0.009 [-0.005, 0.023]	0.005 [-0.010, 0.019]
Share of Voters in Hotspots	-0.025 [-0.142, 0.091]	-0.045 [-0.165, 0.074]
N. Hotspots in Precinct	0.010 [-0.005, 0.025]	0.008 [-0.008, 0.023]
Democrat Share	**-1.387** [-1.784, -0.990]	**-1.189** [-1.596, -0.782]
Prop. Black	0.004 [-0.251, 0.259]	-0.256 [-0.517, 0.006]
Prop. Vacant	-0.018 [-0.244, 0.209]	0.095 [-0.137, 0.327]
Prop. Under 18	-0.114 [-0.459, 0.231]	-0.056 [-0.409, 0.298]
Prop. Single Mother	0.111 [-0.166, 0.389]	0.140 [-0.145, 0.424]
Prop. Poverty	-0.057 [-0.226, 0.112]	-0.006 [-0.179, 0.167]
Prop. Unemployed	-0.019 [-0.252, 0.214]	-0.040 [-0.279, 0.199]
Prop. College	-0.180 [-0.377, 0.018]	-0.037 [-0.239, 0.166]
Prop. Welfare	-0.181 [-0.789, 0.427]	-0.036 [-0.660, 0.587]
Observations	63	63
R^2	0.972	0.977
Adjusted R^2	0.965	0.971
Residual Std. Error (df = 49)	0.049	0.050
F Statistic (df = 13; 49)	131.7	162.7

Note: OLS estimates with 95% confidence intervals. All variables are measured at the precinct level.

H Map of OSS Zone and Hotspots

Figure H.1: **Map of OSS Zone 5 and Experimental Hotspots**



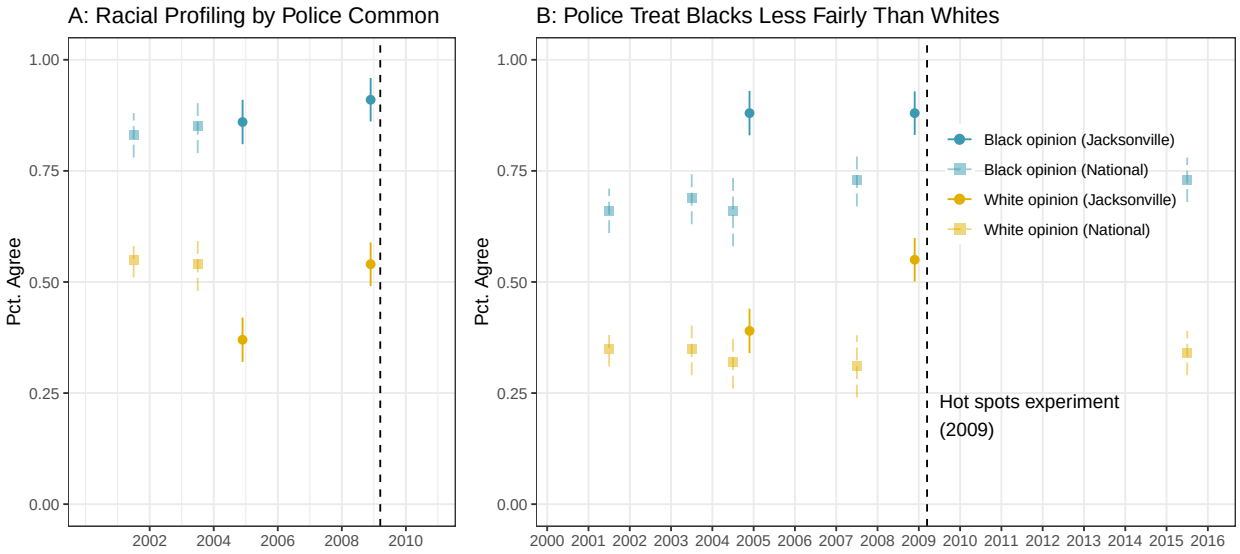
Date Created: December 1, 2008 Jacksonville Sheriff's Office Crime Analysis Unit (JLR)

I Jacksonville Public Opinion Surveys

Survey data on perceived safety comes from the Jacksonville Community Council (JCCI), which has conducted annual "Quality of Life" surveys of the city's residents since 1985, donated by the survey firm *American Public Dialogue*. Each wave of the survey includes between 400 and 500 respondents drawn from a random sample of phone numbers in Duval County, which was stratified to achieve balance on race, gender, and geography. To measure feelings of safety, respondents were given the following prompt: "A feeling of safety is also part of the quality of life. Do you feel safe walking alone at night in your neighborhood?"

I additionally draw on a separate set of surveys of Jacksonville residents fielded in 2004 and 2008 as part of JCCI's "Race Relations Progress Report" series. Both surveys include oversamples of Black residents (n =

Figure I.1: **Public Opinion on Police by Race**



400 and 424, respectively), as well as questions on policing that closely match those asked in Gallup's Annual *Minority Relations Poll*, allowing me to directly compare attitudes by racial group within Jacksonville to the country more broadly. As can be seen in Figure I.1, I find that in 2008, 88% of Black respondents in Jacksonville agreed that local police treated Blacks less fairly than Whites (versus 73% nationally the year previously), and 91% believed racial profiling by the police was widespread (versus 85% nationally). In short, at the time of the experiment Black residents of Jacksonville were at least, if not more, critical of the police as Black Americans elsewhere.

Comparison of question wording:

- JCCI Poll: "Just your impression, are blacks in Jacksonville treated less fairly than whites in dealing with the police?"
- Gallup Poll: "Just your impression, are Black people in your community treated less fairly than White people in the following situations? How about – in dealing with the police, such as traffic incidents?"

J Police Computer-Aided Dispatch Data

J.1 Data Preparation and Cleaning

I obtained incident-level computer aided dispatch (CAD) data covering January 1st, 2007 to May 1st, 2011 from the Jacksonville Sheriff's Office (JSO) after an extensive, 9 month public records request process. Because the JSO would only release incident-level data for specific addresses, all incidents come from the list of addresses lying within the hotspots (plus a 100 foot buffer), which I generated using parcel-level shapefiles

obtained from the Florida Department of Revenue.

The CAD data are stored on two systems. Data from the older system, covering calls prior to 2010, includes a unique incident identifier; the source of the call (911, non-emergency phone line, or an officer); the call location (street address); the basis for the call; whether the call was resolved by the department's Teleserv unit (which handles certain low priority complaints by phone, rather than dispatching patrol officers); the date and time the call was received, responded to, and closed; and the outcome of the call. The newer system, covering 2010 onward, contains less temporal information about each incident (only the date and time the call was received), but includes two additional fields: the ID number of the responding officer, and the priority given to the incident by dispatchers (on a scale of 1 to 6).

I received the data as $\approx 20,000$ pages of PDF image files, which I converted into machine-readable, tabular data using the Mistral Optical Character Recognition API, which is based on a large vision language model (`mistral-ocr-2505`).[24] While the model is closed-source, it was chosen for its ability to handle the complex layout of the data. This process resulted in a dataset of 296,741 calls for service. To verify that the correct number of calls had been extracted from the PDFs, I summed the true counts of calls (which are listed at the end of each PDF document) and cross-checked this total against the OCR output. I find an extremely low error rate — the number of calls is within $\approx 0.01\%$ of the true total.

To further assess the accuracy of the OCR model, I checked the observed values of variables in the data with a known format or range. The results for data from the newer system (which had a more complicated table layout) are summarized in the table below. The results speak to the high accuracy of the OCR model. Not a single date fell outside the known date ranges of the data. For only one variable, signal code, was there more than a handful of invalid values. Nearly all of these discrepancies were caused by characters being incorrectly transcribed as similar looking numbers or letters. For example, the signal "IJ" being written incorrectly as "U" or the letter "O" appearing as the number "0." Once these minor errors were corrected, there were only 51 calls with missing/invalid signal codes (less than 0.1% of the total observations).

Table J.1: **OCR Validation**

Variable	Check	Percent Passed
Call Source	Is equal to "MDT/OFFICER," "PHONE," or "911"	99.9%
Date	Is date (YYYY-MM-DD)	100%
Date	Is in range: 2010-01-01 to 2011-05-01	100%
Priority Rank	Is in range: 1 to 6	100%
Signal Code	Is in set of valid signal codes	97.2%

As a final check, I took two random samples of 100 rows of the data — one from the pre-2010 data, and one

24. Mistral claimed a 96.12% accuracy rate in table recognition/extraction at the time I accessed the OCR API.

from the newer system. Comparing these calls for service to the original images, I found no discrepancies in the key variables of interest: address, date, source, and signal code.

CAD incidents related to certain types of crimes, including sexual assault and child abuse, are not considered public record in the state of Florida and were redacted by the JSO. In the pre-2010 data, the JSO manually redacted the alphanumeric codes detailing the type of incident and the call's outcome, but not the call's date or location. The JSO's newer system can filter incidents by type, so redacted incidents are missing completely. To maintain consistency across the two time frames I drop the partially redacted incidents from the older data.[25]

Because there is no standardized coding system for CAD data, I used the JSO's signal codes and descriptions to group calls into twelve general categories based in part on the categorizations developed by Lum, Koper, and Wu (2022). The first three categories primarily capture officer activity; however, a notable number of these CAD incidents are labeled as originating from 911 or non-emergency calls, particularly cancellations, follow-up investigations, and evidence technician requests. Some of these events are likely to be genuine citizen-initiated calls, while others may be the result of clerical errors. Following Lum, Koper, and Wu (2022), I retain these CAD incidents rather than dropping or reclassifying them. Excluding these calls from my analyses yields similar results.

Officer-Initiated Calls

1. *Administrative:* these calls involve a variety of officer-specific actions including assisting other agencies, consulting with evidence technicians, assisting other officers, and serving warrants, as well as home visits by school resource officers.
2. *Investigation:* this is a catch-all category used by the JSO involving a variety of investigative activities and follow-ups with citizens.
3. *Special Assignment:* this includes proactive problem-solving, hot spots targeting, and door knocks. (Very few incidents were given this label).

Officer or Citizen-Initiated Calls

4. *Disorder:* this includes any call related to social or physical disorder. I follow the commonly used definition of disorder given by Skogan (2012), which encompasses "unsettling or potentially threatening and perhaps unlawful public behaviors," as well as "overt signs of negligence or unchecked decay [and] the visible consequences of malevolent misconduct" (175). This includes noise complaints, illegal parking or dumping, vandalism, fireworks, abandoned vehicles, animals, intoxicated individuals, disputes, drug use, juvenile complaints, fights, obscene or threatening phone calls, and prostitution.
5. *Interpersonal:* this involves non-violent interpersonal incidents including stalking, missing persons, and kidnapping.[26]

25. In several instances the JSO mistakenly sent me both the redacted and unredacted versions of the same documents. Based on this, it appears that nearly all of the redacted incidents were related to sexual assault.

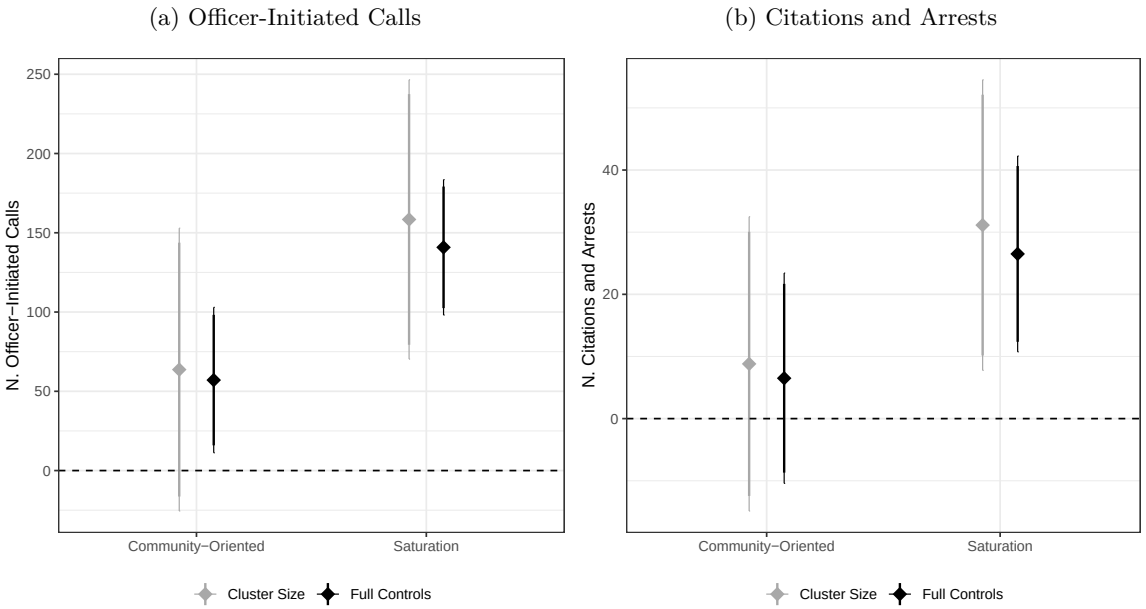
26. While kidnapping is officially classified as a violent crime in some states, the overwhelming majority of kidnapping cases are thought to involve custody disputes over children (Tillyer, Tillyer, and Kelsay 2015). I classify kidnapping as a violent incident only when it was coded as involving a weapon.

6. *Medical:* this includes calls related to medical and mental health emergencies, including cardiac arrest, suicide, or injured persons. While these calls are typically handled by emergency medical services, the calls in the CAD system are those in which a police officer was also dispatched.
7. *Property:* this covers all property-related crimes, including theft, burglary, fraud, identity theft, counterfeits, shoplifting, arson, and damaged property.
8. *Service Requests:* this involves non-crime-related incidents such as requests for information or property checks, assisting a motorist, and reports of downed wires, gas leaks, fires, etc.
9. *Suspicion:* this includes reports of suspicious persons, "prowlers," "peeping toms," mentally ill individuals, and armed individuals not classified as engaging in any particular crime.
10. *Traffic:* this includes all traffic-related incidents including reports of reckless/drunk driving, accidents, and hit-and-runs. For officer-initiated incidents this also includes traffic stops.
11. *Violence:* this includes homicide, fights, assaults, robbery, and home invasions, along with other crimes or potential criminal activity that involve weapons, such as armed burglary, disputes, or kidnapping. While this category includes crimes such as sexual assault, these cases are not considered public record in Florida and were redacted by the JSO.
12. *Other:* this involves miscellaneous non-crime incidents including "strike," "report to hospital," "industrial accident," and "unverified 911 call."

To identify unfounded incidents, as well as those ending in a citation or arrest, I rely on the standardized disposition codes in the data. From the 2010 data onward, these were replaced by text-based descriptions, which I convert into standardized codes using string matching.

J.2 Manipulation Checks

Figure J.1: **Effect of Treatment on Officer Behavior During Experimental Period**

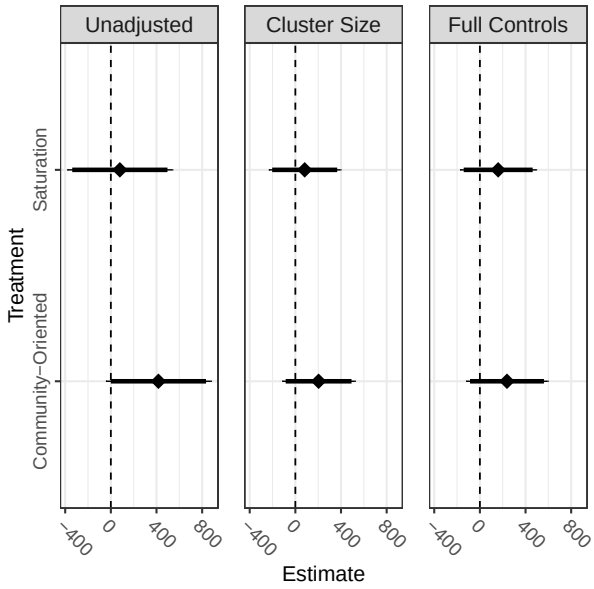


Note: OLS Estimates with 90 and 95% confidence intervals, with a Bonferroni adjustment for four hypothesis tests. All outcomes measured at the hotspot level. Covariate-adjusted estimates control for the count of violence-related calls for service from January 2007 through May, 2008; total population, percent Black, and percent under the age of 18 in 2008; as well as the poverty rate and share of female-headed households in 2000 (US Census Bureau 2000, 2010; JSO, n.d.). Census block-level data is mapped to hotspots using areal interpolation.

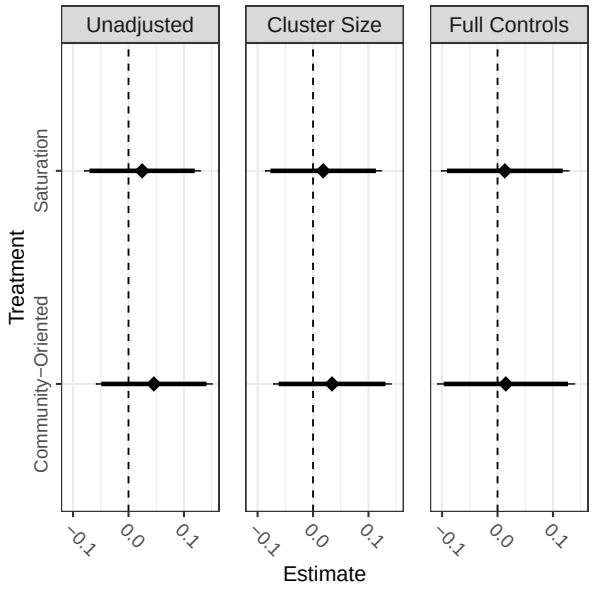
J.3 Post-Treatment Call Patterns

Figure J.2: **Post-Treatment Officer Behavior**

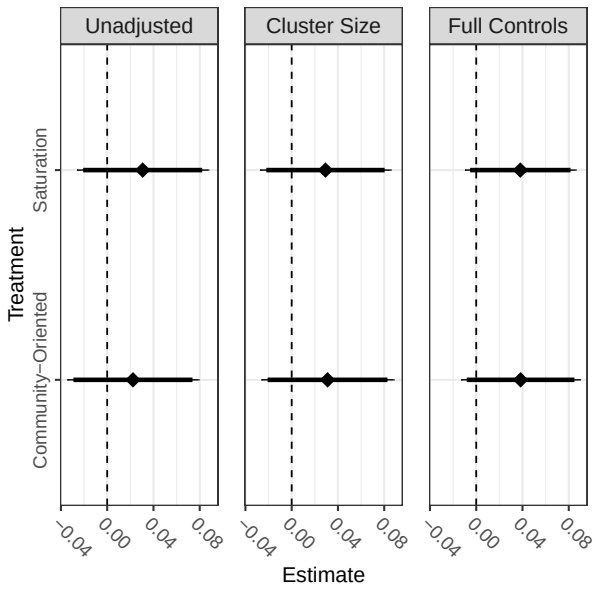
(a) Count of Officer-Initiated Calls



(b) Pct. Calls Initiated by Officers



(c) Pct. Officer Initiated Calls Ending in Citation or Arrest

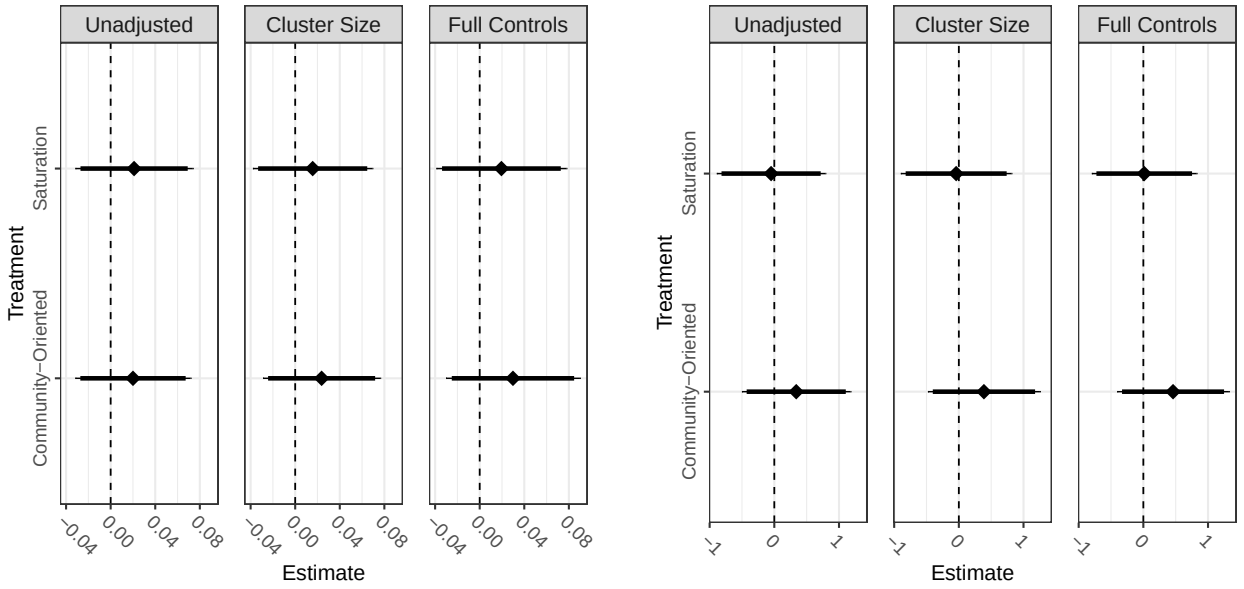


Note: OLS Estimates with 90 and 95% confidence intervals, with a Bonferroni adjustment for six hypothesis tests. All outcomes measured at the hotspot level. Covariate-adjusted estimates include controls for estimated total population, percent Black, and percent under the age of 18 in 2008, as well as the poverty rate and share of female-headed households in 2000 (US Census Bureau 2000, 2010; JSO, n.d.). Census block-level data is mapped to hotspots using areal interpolation.

Figure J.3: **Post-Treatment Citizen-Initiated Calls**

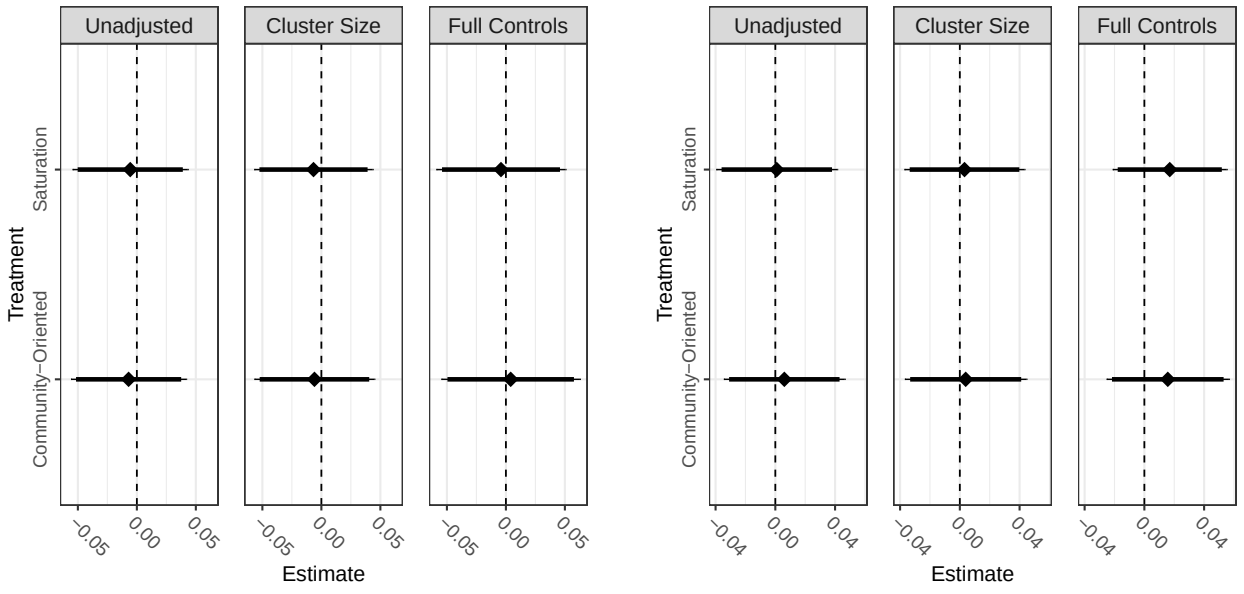
(a) Pct. Low Priority Calls

(b) Ratio of Calls Concerning Violent Crime to those about Suspicious Activity



(c) Pct. Calls Diverted to Teleserv Unit

(d) Pct. Nonverified Calls



Note: OLS Estimates with 90 and 95% confidence intervals, with a Bonferroni adjustment for eight hypothesis tests. All outcomes measured at the hotspot level. Covariate-adjusted estimates include controls for estimated total population, percent Black, and percent under the age of 18 in 2008, as well as the poverty rate and share of female-headed households in 2000 (US Census Bureau 2000, 2010; JSO, n.d.). Census block-level data is mapped to hotspots using areal interpolation.

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